Existence of locally correct interleavings between merge trees

Thijs Beurskens, Tim Ophelders, Bettina Speckmann, Kevin Verbeek

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Editorial scope. The constructive proof maintains dominant partial interleavings while adding minimal augmentations using realizing critical pairs, preserving local correctness over arbitrary restrictions and terminating through newly specified vertices. Complete residual-distance and bottleneck geometry is included. Complete author full-version mathematics under own CCBY4. No support split or software rerun.

Allowed standard prerequisites (editorial).

- Merge-tree interleavings and bottleneck critical values. Original author definitions/references retained; all source-specific partial-map, residual extension, critical-pair and augmentation proofs included.

1 Introduction

Terrains that vary over time are a common data type in various application areas, such as scientific computing or geographic information science. Typically such data sets are very large; to analyze them efficiently one often needs a compact abstraction that captures salient features. Topological data analysis (TDA) provides several topological descriptors [15], such as persistence diagrams, Reeb graphs, and Morse-Smale complexes, that can serve as abstractions for 2D, 3D, or higher-dimensional terrains. In this paper we focus in particular on merge trees, which are graph-based descriptors that encode the evolution of connected

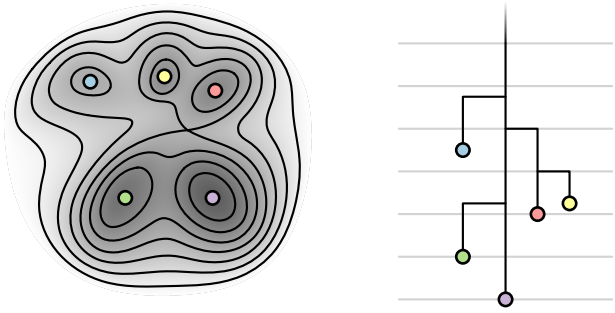


Figure 1: A terrain and its merge tree; leaves and internal vertices represent minima and saddles.

components of sublevel or superlevel sets of a terrain (see Figure 1). To analyze temporal sequences of merge trees one does not only need a compact representation, but also efficient ways to compare and match them over time. For merge trees in particular, several distance measures have been proposed, including the edit distance [12], the Wasserstein distance [10], and the interleaving distance [7]. However, a distance measure returns only a numeric value that quantifies the similarity of two merge trees T_1 and T_2. In many application scenarios, we would also like to track the merge trees over time and hence need to understand which parts of T_1 most closely correspond to parts of T_2. That is, we are interested in a matching between T_1 and T_2. The interleaving distance in fact relies on a type of matching between the two trees, the so-called interleaving. An interleaving is a pair of ancestor-preserving maps, one from T_1 to T_2 and one from T_2 to T_1, whose compositions send points to ancestors: if a point x of T_1 is mapped to a point y of T_2, then y must be mapped to an ancestor of x, and vice versa. The interleaving distance is the maximum shift—the distance between a point and its image—minimized over all interleavings between the trees. We give exact definitions of the interleaving distance and interleavings in Sections 2 and 3. Two merge trees typically admit many optimal interleavings. The interleaving distance is a bottleneck measure and hence a given optimal interleaving might be locally quite "loose": the optimal shift might exceed what is needed and thereby fail to capture local similarities between the trees; see Figure 2. Inspired by the concept of locally correct Fréchet matchings for curves [3], in the

following we are hence developing a notion of locally optimal interleavings which allows us to distinguish between “tight” and “loose” optimal interleavings.

Contributions. A locally correct Fréchet matching is optimal even when considering any two matched subcurves. This is a desirable feature, however, it cannot be directly reproduced using interleavings. First of all, there might not even be a one-to-one correspondence between the points of T_1 and T_2. Furthermore, two optimal interleavings between different pairs of subtrees do not necessarily combine into an optimal interleaving between the two complete trees. As a first step we hence introduce a novel generalization of the interleaving distance that can handle additional constraints on the maps. Specifically, we aim to use constraints that force a point x to be mapped to an ancestor of a particular point y within the other tree. Over interleavings that satisfy such constraints, we aim to minimize the maximum shift over points x that are either unconstrained or map to a strict ancestor of their corresponding point y. We refer to the resulting distance as the residual interleaving distance. In Section 3, we provide formal definitions and show that an optimal interleaving exists for any constraints. In Section 4 we then propose a definition for locally correct interleavings that builds upon the residual interleaving distance. Specifically, we define a restriction of an interleaving $\mathcal{I} = (\alpha, \beta)$ to two subsets S_1 of T_1 and S_2 of T_2 as the restriction of α to S_1 together with the restriction of β to S_2. We call $\mathcal{I}$ locally correct if it minimizes the residual interleaving distance

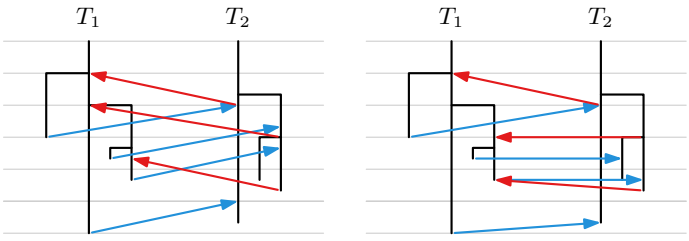


Figure 2: Two optimal interleavings between merge trees T_1 and T_2. Intuitively, the right interleaving induces a tighter and more meaningful matching than the left interleaving.

with respect to any restriction of $\mathcal{I}$. We constructively prove that a locally correct interleaving always exists.

Related work. The interleaving distance between merge trees was defined by Morozov et al. [7] as a tool to study the stability of merge trees. Agarwal et al. [1] establish a connection between the interleaving distance and the Gromov-Hausdorff distance, and they argue that approximating the interleaving distance within a factor of 3 is NP-hard. They also describe an $\mathcal{O}(\min(n, \sqrt{rn}))$-approximation algorithm, where n is the total size of the trees and r is the ratio of the longest to the shortest edge. Touli and Wang [13] provide an equivalent definition of the interleaving distance in terms of a single map and use this characterization to design an FPT-algorithm for computing the interleaving distance exactly. More recently, Gasparovic et al. [5] show that the interleaving distance can be formulated in terms of a related metric [8] for labeled merge trees, also known as phylogenetic trees. Several attempts have been made to develop versions of the interleaving distance that can be used in practical applications. In particular, the work by Gasparovic et al. [5] gave rise to heuristic algorithms for computing the interleaving distance [4] and for comparing terrains [14, 16]. Pegoraro [9] uses yet another reformulation of the interleaving distance, in terms of couplings, as the basis of a heuristic for computing the distance via linear integer programming. Lastly, Beurskens et al. [2] introduce a variant of the interleaving distance that imposes total orders on the merge trees, which can be computed efficiently via a connection to the Fréchet distance between one-dimensional curves.

2 Preliminaries

A merge tree is a pair (T, f), where T is a finite rooted tree, and $f\colon T \to \mathbb{R} \cup \{\infty\}$ is a continuous height function on T that is strictly increasing towards the root and such that the root has height ∞. We identify T with its topological realization: the topological space where each edge of T is represented by a unit segment and connections are according to the adjacencies of T. We

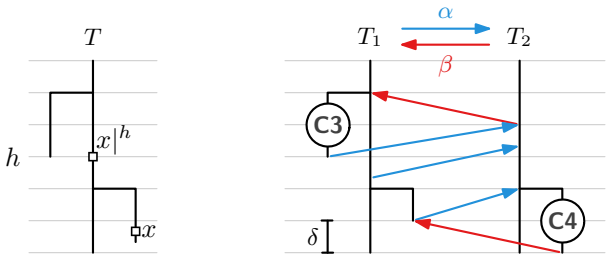


Figure 3: Left: a merge tree T; the point $x|^h$ is the ancestor of x at height h. Right: parts of a pair of δ-compatible maps.

distinguish elements of the topological realization T, which we refer to as points of T, from vertices of the combinatorial tree, which we denote by $V(T)$. When clear from context, we simply write T to refer to the pair (T, f). For two points x_1, x_2 of T, we say x_1 is a descendant of x_2, written $x_1 \preceq x_2$, if there exists an f-monotonically increasing path in T from x_1 to x_2. If furthermore $x_1 \neq x_2$ then x_1 is a strict descendant of x_2. We say x_2 is a (strict) ancestor of x_1 if x_1 is a (strict) descendant of x_2. Lastly, for a point x and a height value $h \geq f(x)$, we use $x|^h$ to denote the unique ancestor of x at height $f(x|^h) = h$ (see Figure 3 left).

Compatible maps. The original definition of the interleaving distance between two merge trees (T_1, f_1) and (T_2, f_2) was given by Morozov et al. [7].

Definition 1 (Morozov et al. [7])**.** Fix a value $\delta \geq 0$. Two continuous maps $\alpha\colon T_1 \to T_2$ and $\beta\colon T_2 \to T_1$ are δ-compatible between T_1 and T_2 if for all $x \in T_1$ and all $y \in T_2$:

(C1) $f_2(\alpha(x)) = f_1(x) + \delta$,

(C2) $f_1(\beta(y)) = f_2(y) + \delta$,

(C3) $\beta(\alpha(x)) = x|^{f_1(x)+2\delta}$, and

(C4) $\alpha(\beta(y)) = y|^{f_2(y)+2\delta}$.

The interleaving distance $d(T_1, T_2)$ between T_1 and T_2 is the infimum δ for which there exist δ-compatible maps (see Figure 3 right).

Critical values. Both Agarwal et al. [1] and Touli and Wang [13] have shown that the interleaving distance between any

two merge trees attains a value in a set of critical values. Specifically, let $\Delta \coloneqq \Delta(T_1, T_2)$ be the union of three sets Δ_1, Δ_2, and Δ_3, with

$$\begin{aligned} \Delta_1 &\coloneqq \{|f_1(v) - f_2(w)| : v \in V(T_1), w \in V(T_2)\}, \\ \Delta_2 &\coloneqq \{\tfrac{1}{2}|f_1(v_1) - f_1(v_2)| : v_1, v_2 \in V(T_1)\}, \\ \Delta_3 &\coloneqq \{\tfrac{1}{2}|f_2(w_1) - f_2(w_2)| : w_1, w_2 \in V(T_2)\}. \end{aligned} \tag{1}$$

Lemma 2 ([1, 13])**.** For any two merge trees T_1 and T_2, it holds that $d(T_1, T_2) \in \Delta(T_1, T_2)$.

3 Generalizing the Interleaving Distance

The interleaving distance is based on a pair of maps. As noted by Agarwal et al. [1], we can relax the requirements on these maps to allow varying height differences between points and their images. Building on this relaxed definition, we introduce a generalized version of the interleaving distance that accommodates constraints on the maps we consider. Specifically, each constraint specifies that a point x must be mapped to an ancestor of a particular point y within the other tree. Our goal is then to find the optimal pair of maps that "extends" a given set of constraints. We call the corresponding distance the residual interleaving distance. To define it formally, we first introduce partial up-maps, which represent the allowed maps between the two merge trees. In Section 3.1 we relax Definition 1 using partial up-maps. In Section 3.2 we then give the formal definition of the residual interleaving distance.

Partial up-maps. Let (T_1, f_1) and (T_2, f_2) be two merge trees. An arrow from T_1 to T_2 is a pair of points $(x, y) \in T_1 \times T_2$ such that $f_1(x) \leq f_2(y)$. The shift of an arrow (x, y) is the height difference between x and y, denoted $\sigma(x, y) \coloneqq f_2(y) - f_1(x)$. Consider a subset of points $S \subseteq T_1$. A map $\varphi\colon S \to T_2$ is a partial up-map from S to T_2 if (i) $f_2(\varphi(x)) \geq f_1(x)$ for all points x of S and (ii) it preserves ancestors, that is, $x_1 \preceq x_2$ implies $\varphi(x_1) \preceq \varphi(x_2)$ for all points x_1, x_2 of S. We use $\mathrm{dom}(\varphi) \coloneqq S$ to denote the domain of φ.

For each point x of S, the pair $(x, \varphi(x))$ is an arrow. We denote the set of arrows of φ by $A_\varphi \coloneqq \{(x, \varphi(x)) : x \in S\}$. We define the shift of the partial up-map φ as the supremum shift over all arrows of φ:

$$\sigma(\varphi) \coloneqq \sup\{\sigma(a) : a \in A_\varphi\}. \tag{2}$$

Given an arrow (x, y) from S to T_2, we say φ extends (x, y) if $\varphi(x) \succeq y$, and we say φ uses (x, y) if $\varphi(x) = y$. Similarly, for a subset $S' \subseteq S$ and a partial up-map φ' from S' to T_2, we say φ extends or uses φ' if, respectively, φ extends or uses all arrows of φ' (Figure 4 left). For any $\delta \geq 0$, we can define a partial up-map $\varphi^{\uparrow}[\delta]$ from S to T_2 that extends φ and such that the shift of each arrow is at least δ. Specifically, for each point x of S, let $h_x \coloneqq \max(f_1(x) + \delta, f_2(\varphi(x)))$ and define $\varphi^{\uparrow}[\delta](x) \coloneqq \varphi(x)|^{h_x}$ (see Figure 4 middle).

Lemma 3. For any $\delta \geq 0$, the map $\varphi^{\uparrow}[\delta]$ is a partial up-map that extends φ.

Proof. Let $\varphi' \coloneqq \varphi^{\uparrow}[\delta]$. By construction, we have $f_2(\varphi'(x)) \geq f_1(x)$. To see that φ' preserves ancestors, take two points x_1, x_2 of S such that x_2 is an ancestor of x_1. Since φ preserves ancestors, $\varphi(x_2)$ is an ancestor of $\varphi(x_1)$. If $\varphi'(x_1) = \varphi(x_1)$, we get $\varphi'(x_1) \preceq \varphi(x_2) \preceq \varphi'(x_2)$. Otherwise $\varphi'(x_1)$ is a strict ancestor of $\varphi(x_1)$. By construction, $f_2(\varphi'(x_1)) = f_1(x_1) + \delta \leq f_1(x_2) + \delta \leq f_2(\varphi'(x_2))$. Both $\varphi'(x_1)$ and $\varphi'(x_2)$ are ancestors of $\varphi(x_1)$, so $\varphi'(x_2)$ is an ancestor of $\varphi'(x_1)$. So φ' is a partial up-map, and by construction also extends φ. □

We define arrows and up-maps from a subset S of T_2 to T_1 symmetrically.

Partial interleavings. Fix two subsets $S_1 \subseteq T_1$ and $S_2 \subseteq T_2$. A partial interleaving $\mathcal{P}$ between S_1 and S_2 is a pair of partial up-maps $\varphi\colon S_1 \to T_2$ and $\psi\colon S_2 \to T_1$ such that for all points x of S_1 and all points y of S_2 it holds that if $\varphi(x) \preceq y$ then $\psi(y) \succeq x$, and if $\psi(y) \preceq x$ then $\varphi(x) \succeq y$ (Figure 4 right). The shift of $\mathcal{P}$, denoted $\sigma(\mathcal{P})$, is the maximum of the shift of φ and the shift of ψ. We define the arrows of $\mathcal{P}$ as the union of the arrows of φ and the arrows of ψ, denoted $A_\mathcal{P} \coloneqq A_\varphi \cup A_\psi$. Given another partial interleaving $\mathcal{P}' = (\varphi', \psi')$, we say $\mathcal{P}$ extends $\mathcal{P}'$ if φ extends φ' and ψ extends ψ'. Similarly, we say $\mathcal{P}$ uses $\mathcal{P}'$ if φ uses φ' and ψ uses ψ'. For any $\delta \geq 0$, the pair $(\varphi^{\uparrow}[\delta], \psi^{\uparrow}[\delta])$ is a partial interleaving.

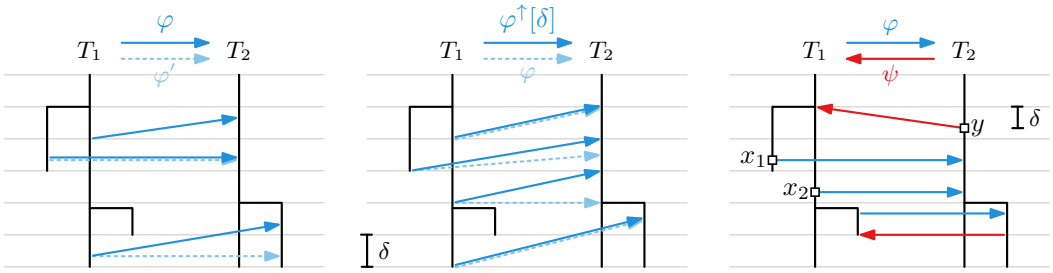


Figure 4: Left: a partial up-map φ that extends another partial up-map φ'. Middle: a partial up-map φ and the corresponding map $\varphi^\uparrow[\delta]$; each non-dashed arrow has shift at least δ. Right: a partial interleaving with shift δ; note that $\psi(y)$ is an ancestor of both x_1 and x_2.

Lemma 4. For any $\delta \geq 0$, the pair $(\varphi^\uparrow[\delta], \psi^\uparrow[\delta])$ is a partial interleaving that extends (φ, ψ).

Proof. Denote $\varphi' \coloneqq \varphi^\uparrow[\delta]$ and $\psi' \coloneqq \psi^\uparrow[\delta]$. By Lemma 3, the maps φ' and ψ' are partial up-maps that extend φ and ψ, respectively. Therefore, it suffices to show that the pair (φ', ψ') is a partial interleaving. Let x be a point of S_1 and y be a point of S_2 such that $\varphi'(x) \preceq y$. By construction, the point $\varphi'(x)$ is an ancestor of $\varphi(x)$, so y is also an ancestor of $\varphi(x)$. Since (φ, ψ) is a partial interleaving, the point $\psi(y)$ must be ancestor of x. As $\psi'(y)$ is an ancestor of $\psi(y)$, we get $\psi'(y) \succeq x$. Symmetrically, if $\psi'(y) \preceq x$ then $\varphi'(x) \succeq y$. □

For a partial interleaving $\mathcal{P} = (\varphi, \psi)$, we use $\mathcal{P}.\varphi$ and $\mathcal{P}.\psi$ to denote φ and ψ, respectively.

3.1 Complete Interleavings

If $S_1 = T_1$, we call a partial up-map $\alpha \colon S_1 \to T_2$ a complete up-map, or simply an up-map, from T_1 to T_2. Symmetrically, if $S_2 = T_2$, we call a partial up-map $\beta \colon S_2 \to T_1$ a (complete) up-map from T_2 to T_1. Correspondingly, if $S_1 = T_1$ and $S_2 = T_2$, we call the partial interleaving (α, β) a (complete) interleaving. The defining property for a complete interleaving then becomes that for all $x \in T_1$ it holds that $\beta(\alpha(x)) \succeq x$ and for all $y \in T_2$ it holds that $\alpha(\beta(y)) \succeq x$. In Lemma 7, we show that the interleaving distance is equal to the

infimum δ for which T_1 and T_2 admit an interleaving with shift δ. First, it is not hard to see that any pair of δ-compatible maps form an interleaving with shift δ.

Observation 5. Any pair of δ-compatible maps is an interleaving with shift δ.

Conversely, for a complete interleaving (α, β) with shift δ, we show that there always exists a pair of δ-compatible maps. In particular, the pair $(\alpha^{\uparrow}[\delta], \beta^{\uparrow}[\delta])$ is δ-compatible.

Lemma 6. For any interleaving (α, β) with shift δ, $\alpha^{\uparrow}[\delta]$ and $\beta^{\uparrow}[\delta]$ are δ-compatible maps.

Proof. Denote $\alpha' \coloneqq \alpha^{\uparrow}[\delta]$ and $\beta' \coloneqq \beta^{\uparrow}[\delta]$. We first show that α' is continuous; proving that β' is continuous is analogous. Let x be a point of T_1 and consider a small neighborhood X of x in which each point x' is reachable from x by an f_1-monotone path. To prove continuity of α' at x, we show that there exists an f_2-monotone path between $\alpha'(x)$ and $\alpha'(x')$ and that the height difference between $\alpha'(x)$ and $\alpha'(x')$ is bounded. By construction and choice of δ, it follows that α' maps each point of T_1 to a point of T_2 exactly δ higher. In particular, we obtain $|f_2(\alpha'(x')) - f_2(\alpha'(x))| = |f_1(x') - f_1(x)|$. As α' is an up-map, we know that $x \preceq x'$ implies $\alpha'(x) \preceq \alpha'(x')$, and that $x' \preceq x$ implies $\alpha'(x') \preceq \alpha'(x)$. In other words, there exists an f_2-monotone path between $\alpha'(x)$ and $\alpha'(x')$. So α' is continuous. Lastly, as (α', β') is an interleaving, and the shift of each arrow is exactly δ, it follows that $\beta'(\alpha'(x)) = x|^{f_1(x)+2\delta}$ for all $x \in T_1$, and $\alpha'(\beta'(y)) = y|^{f_2(y)+2\delta}$ for all $y \in T_2$. □

Combining Observation 5 and Lemma 6 we obtain Lemma 7.

Lemma 7. $d(T_1, T_2) = \inf\{\delta : T_1 \text{ and } T_2 \text{ admit a complete interleaving with shift } \delta\}$.

Optimal interleavings. The interleaving distance is defined as an infimum, but both Gasparovic et al. [6] and Pegoraro [9] proved that the infimum can be replaced by a minimum: there always exists an interleaving with shift $d(T_1, T_2)$ (Theorem 9). We call an interleaving that realizes the interleaving distance an optimal interleaving. Both proofs rely on alternative definitions of the interleaving distance, namely in terms of labellings [6] and in

terms of couplings [9]. We now give a more direct proof of Theorem 9. To do so, we first show that for an interleaving whose shift is not equal to a critical value, there exists an interleaving whose shift is strictly smaller.

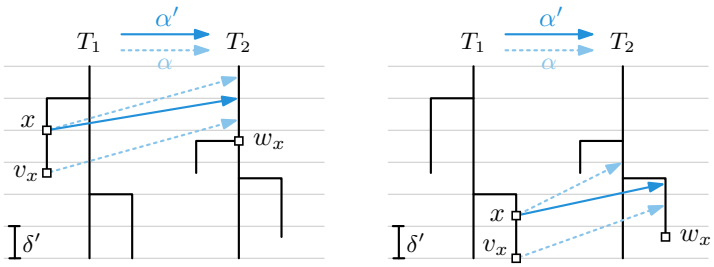


Figure 5: Illustration of the map α'. The image of each point x is "pushed down" to the correct height. On the left there is only one option, but on the right we need to choose the correct subtree.

Lemma 8. For any complete interleaving $\mathcal{I} = (\alpha, \beta)$ with shift $\delta \notin \Delta$, there exists another interleaving $\mathcal{I}'$ whose shift is equal to a critical value that is strictly less than δ.

Proof. Let δ' be the greatest value in $\Delta(T_1, T_2)$ that satisfies $\delta' \leq \delta$. As there are only finitely many critical values, there is such a value δ'. W.l.o.g., we assume that all arrows of $\mathcal{I}$ have shift at least δ'. Otherwise, we can replace α and β with $\alpha^{\uparrow}[\delta']$ and $\beta^{\uparrow}[\delta']$, respectively. For each point x of T_1, let $v_x \in V(T_1)$ be the lower endpoint of the edge that contains x; if x is a vertex then $v_x = x$. Moreover, let w_x be the lower endpoint of the edge that contains $\alpha(v_x)$. The value $f_2(w_x) - f_1(v_x)$ is a critical value, so by choice of δ' we know that $f_2(w_x) - f_1(v_x) \leq \delta'$. In other words, $f_2(w_x) \leq f_1(x) + \delta'$. Therefore, we can safely define $\alpha'(x) = w_x|^{f_1(x)+\delta'}$ We define the map β' from T_2 to T_1 symmetrically. See Figure 5. We first argue that α' is an up-map; the proof for β' is analogous. Let x_1 and x_2 be points of T_1 such that $x_1 \preceq x_2$. If $v_{x_1} = v_{x_2}$ then we directly obtain $\alpha'(x_1) \preceq \alpha'(x_2)$. Otherwise, we must have $v_{x_1} \preceq x_1 \preceq v_{x_2}$. Since α preserves ancestors, it follows that $\alpha(v_{x_1}) \preceq \alpha(v_{x_2})$ and thus $w_{x_1} \preceq w_{x_2}$. By construction, we obtain $\alpha'(x_1) \preceq \alpha'(x_2)$. So indeed, α' preserves ancestors. We additionally claim that $\alpha'(x) \preceq \alpha(x)$ for all x. We know that $\alpha(x) \succeq \alpha(v_x) \succeq w_x$, so the point w_x is a descendant of both $\alpha(x)$ and $\alpha'(x)$. By construction, it holds that $f_2(\alpha'(x)) \leq \delta' \leq f_2(\alpha(x))$. Our claim follows. It remains to show that (α', β') is an interleaving. We first show that $\beta'(\alpha'(v)) \succeq v$ for all vertices v of T_1. Let v' be the lower endpoint of the edge that contains $\beta(\alpha(v))$. Then we know that the value $\delta^* := \frac{1}{2}(f_1(v') - f_1(v))$ is a critical value that satisfies $\delta^* \leq \delta$. By choice of δ' it then

follows that $\delta^* \leq \delta'$. Let $x \in T_1$ For all points y of T_2, we have $\beta'(y) \preceq \beta(y)$. Taking $y = \alpha'(x)$, we get $\beta'(\alpha'(x)) \preceq \beta(\alpha'(x))$. Similarly, we have $\alpha'(x) \preceq \alpha(x)$. Since β preserves ancestors, it follows that $\beta(\alpha'(x)) \preceq \beta(\alpha(x))$. Combining, we get $\beta'(\alpha'(x)) \preceq \beta(\alpha(x))$ for all $x \in T_1$. In particular, for $x = v$, we obtain $\beta'(\alpha'(v)) \preceq \beta(\alpha(v))$. Moreover, by construction we know that the height of $\beta'(\alpha'(v))$ is exactly $f_1(v)+2\delta' \geq f_1(v)+2\delta^*$. It follows that $\beta'(\alpha'(v))$ is an ancestor of v' and thus also of v. Now, for an arbitrary point x of T_1, consider the lower endpoint v_x of the edge that contains x. Both α' and β' preserve ancestors, so we get $\beta'(\alpha'(x)) \succeq \beta'(\alpha'(v_x)) \succeq v_x$. It follows that $\beta'(\alpha'(x)) \succeq x$. An analogous argument shows that $\alpha'(\beta'(y)) \succeq y$ for all $y \in T_2$. □

We now argue that an optimal interleaving always exists. Let $\delta = d(T_1, T_2)$, and let $\delta_1, \delta_2 \in \Delta(T_1, T_2)$ be the greatest value that satisfies $\delta_1 \leq \delta$ and the smallest value that satisfies $\delta_2 > \delta$, respectively. By Lemma 7 we know that for all $\varepsilon > 0$ there is an interleaving with shift $\delta + \varepsilon$. In particular, for ε small enough, there is an interleaving $\mathcal{I}$ with shift $\delta + \varepsilon \in [\delta_1, \delta_2]$. If $\mathcal{I}$ is optimal, we are done. Otherwise, by Lemma 8 it follows that there exists an interleaving with shift at most δ_1, which means that $\delta_1 = \delta$. Theorem 9 directly follows.

Theorem 9. Any two merge trees T_1 and T_2 admit an optimal interleaving.

3.2 Residual Interleaving Distance

Fix a partial interleaving $\mathcal{P} = (\varphi, \psi)$. We refer to a (partial) interleaving that extends $\mathcal{P}$ as a (partial) $\mathcal{P}$-extension. We first observe that there always exists a complete $\mathcal{P}$-extension. Indeed, the maps that take every point of T_1 to the root at infinity of T_2, and every point of T_2 to the root at ∞ of T_1 form a complete interleaving that extends $\mathcal{P}$.

Observation 10. Any partial interleaving $\mathcal{P}$ admits a complete $\mathcal{P}$-extension.

Our goal is to find a "tightest possible" complete $\mathcal{P}$-extension, that is, an interleaving that extends $\mathcal{P}$ and minimizes the shift of the remaining arrows. For some intuition, consider the example in Figure 6. The map φ consists of a single arrow (x, y). Intuitively, the

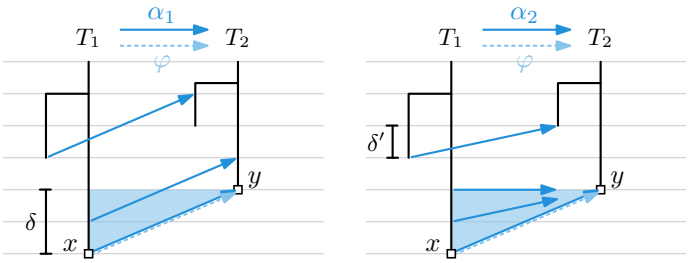


Figure 6: Illustration of the residual shift. The shifts of arrows within the fan $F[(x, y)]$ (represented by the shaded area) are disregarded, so the φ-shift of α_2 is less than the φ-shift of α_1.

map α_2 describes a "tighter" extension of φ than the map α_1. However, if we only disregard the shift of the arrow (x, y), the shift of the remaining arrows of α_1 is equal to the (supremum) shift of the remaining arrows of α_2. The arrow (x, y) implicitly induces a "fan" of arrows that all valid up-maps need to extend. Specifically, any ancestor of x needs to be mapped to an ancestor of y. To distinguish between the maps α_1 and α_2, we therefore not only disregard the shift of (x, y), but of any arrow that is "contained" within such a fan. Formally, we define the fan of an arrow (x, y) as the set of arrows (x', y) for all ancestors x' of x with height at most that of y, written,

$$F[(x, y)] \coloneqq \{(x', y) : x' \in T_1 \text{ such that } x' \succeq x \text{ and } f_1(x') \leq f_2(y)\}.$$

The fan of a collection of arrows A, denoted $F[A]$, is the union of all fans $F[a]$ for $a \in A$. We use fans to define the φ-residual shift, or φ-shift for short. Recall that we use A_φ to denote the set of all arrows $(x, \varphi(x))$. For an arrow a from T_1 to T_2, the φ-residual shift of a, denoted $\sigma_\varphi(a)$, is zero if a lies within $F[A_\varphi]$, and is $\sigma(a)$ otherwise. Similar to the definition of shift in Eq. 2, we define the φ-residual shift of an up-map φ', denoted $\sigma_\varphi(\varphi')$, as the supremum φ-residual shift over all arrows of φ'. That is,

$$\sigma_\varphi(x, y) \coloneqq \begin{cases} 0 & \text{if } (x, y) \in F[A_\varphi], \\ \sigma(x, y) & \text{else,} \end{cases} \qquad \sigma_\varphi(\varphi') \coloneqq \sup\{\sigma_\varphi(a) : a \in A_{\varphi'}\}.$$

Consider again the example in Figure 6. We disregard the shifts of all arrows within the fan $F[(x, y)]$. As a result, the φ-shift of α_2

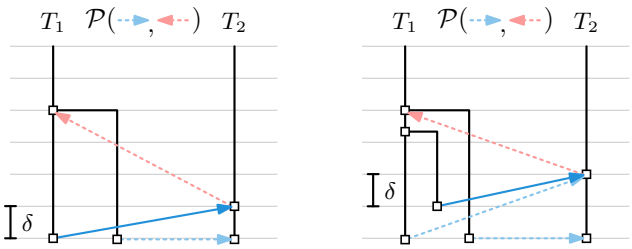


Figure 7: In both examples, the $\mathcal{P}$-residual interleaving distance is determined by the non-dashed, dark blue arrow. The shift of those arrows is equal to δ, which in both cases is not a critical value for T_1 and T_2. In both cases, however, δ is a $\mathcal{P}$-critical value. All critical points are marked.

is δ', whereas the φ-shift of α_1 is δ. We define the ψ-(residual) shift of a partial up-map ψ' symmetrically. Finally, for a partial interleaving $\mathcal{P}' = (\varphi', \psi')$, we define the $\mathcal{P}$-residual shift, or $\mathcal{P}$-shift, of $\mathcal{P}'$ as the maximum of the φ-shift of φ' and the ψ-shift of ψ', denoted $\sigma_{\mathcal{P}}(\mathcal{P}')$.

Definition 11. The $\mathcal{P}$-residual interleaving distance $d_{\mathcal{P}}(T_1, T_2)$ between T_1 and T_2 is the infimum δ for which there exists a complete $\mathcal{P}$-extension whose $\mathcal{P}$-residual shift is δ.

When they are clear from context, we omit the arguments T_1 and T_2 and simply write $d_{\mathcal{P}}$ to denote the $\mathcal{P}$-residual interleaving distance between T_1 and T_2. We say an extension is optimal if its residual shift is equal to the residual interleaving distance.

Critical values and pairs. For the remainder of this section, we assume that $\mathcal{P}$ is finite: both $\mathrm{dom}(\varphi)$ and $\mathrm{dom}(\psi)$ consist of finitely many points. Recall from Section 2 that the non-residual interleaving distance always attains a value in the set of critical values $\Delta := \Delta(T_1, T_2)$. For the residual interleaving distance, this set Δ is not always sufficient (see Figure 7). Therefore, we extend Δ to a set of $\mathcal{P}$-critical values. Specifically, we "mark" all vertices of T_1 and T_2, and all points x of T_1 and y of T_2 that appear in an arrow (x, y) or (y, x) of $\mathcal{P}$. We call these the $\mathcal{P}$-critical points of T_1 and T_2, denoted $C_1[\mathcal{P}]$ and $C_2[\mathcal{P}]$ respectively (see Figure 7). Since $\mathcal{P}$ is finite, there are only finitely many critical points. We

define a set $\Delta_1[\mathcal{P}]$ that contains all height differences between critical points:

$$\Delta_1[\mathcal{P}] \coloneqq \{|f_1(v) - f_2(w)| : v \in C_1[\mathcal{P}], w \in C_2[\mathcal{P}]\}.$$

We define the set of $\mathcal{P}$-critical values as the set $\Delta[\mathcal{P}](T_1, T_2) \coloneqq \Delta_1[\mathcal{P}] \cup \Delta_2 \cup \Delta_3$. When clear from context, we simply write $\Delta[\mathcal{P}]$. Each critical value corresponds to a pair of critical points; we refer to such a pair b as a $\mathcal{P}$-critical pair. We distinguish two types; see Figure 8.

- If $b = (v, w)$ for a point $v \in C_1[\mathcal{P}]$ and a point $w \in C_2[\mathcal{P}]$, then b is an arrow critical pair that corresponds to the critical value $\sigma(v, w)$. We say a partial interleaving (φ, ψ) uses b if $\varphi(v) = w$. The case that $b = (w, v)$ is symmetric.
- If $b = (v_1, v_2)$ for two vertices v_1, v_2 in $V(T_1)$, then b is a zigzag critical pair that corresponds to the critical value $\frac{1}{2}(f_1(v_2) - f_1(v_1))$. We say a partial interleaving (φ, ψ) uses b if there is a point $y \in T_2$ with $f_2(y) = f_1(v_1) + \frac{1}{2}(f_1(v_2) - f_1(v_1))$ such that $\varphi(v_1) = y$ and $\psi(y) = v_2$. The case that $b = (w_1, w_2)$ for two vertices w_1, w_2 in $V(T_2)$ is symmetric.

Similar to Lemma 2, we show that the $\mathcal{P}$-residual interleaving distance always attains a value in the set of $\mathcal{P}$-critical values. We additionally show that there always exists an optimal $\mathcal{P}$-extension and that any such extension uses a realizing critical pair: a critical pair that corresponds to the interleaving distance. We first prove a generalization of Lemma 8:

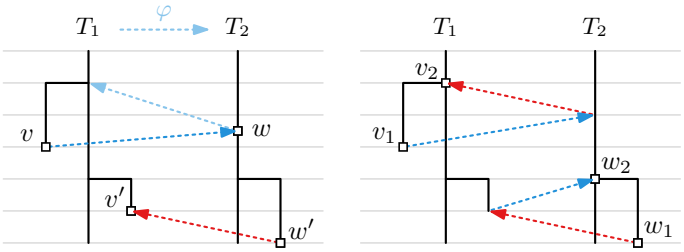


Figure 8: There are two types of critical pairs: arrow critical pairs (v, w) and (w', v') (left) and zigzag critical pairs (v_1, v_2) and (w_1, w_2) (right).

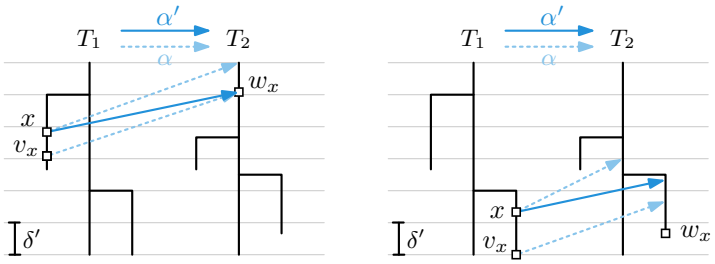


Figure 9: The construction of α'; we need to be careful around the critical points v_x and w_x.

Lemma 12. Let $\mathcal{P}$ be a finite interleaving. For any complete $\mathcal{P}$-extension with $\mathcal{P}$-shift δ that does not use a critical pair that corresponds to δ, there exists another $\mathcal{P}$-extension whose $\mathcal{P}$-shift is equal to a $\mathcal{P}$-critical value that is strictly less than δ.

Proof. Consider a $\mathcal{P}$-extension $\mathcal{I} = (\alpha, \beta)$ with $\mathcal{P}$-shift δ, and assume $\mathcal{P}$ does not use a critical pair that corresponds to δ. We modify $\mathcal{I}$ into an extension $\mathcal{I}' = (\alpha', \beta')$ whose $\mathcal{P}$-shift is equal to a $\mathcal{P}$-critical value that is strictly less than δ. Let $\delta' \in \Delta[\mathcal{P}]$ be the greatest value that satisfies $\delta' < \delta$. We describe the construction of α'; the construction of β' is symmetric. We assume, without loss of generality, that each arrow of α has shift at least δ'. Otherwise, we replace α by $\alpha^{\uparrow}[\delta']$. Since the residual shift of an arrow is bounded by its shift, and since $\delta' < \delta$, the φ-residual shift of $\alpha^{\uparrow}[\delta']$ is bounded by δ. For $x \in T_1$, let $v_x \in C_1[\mathcal{P}]$ be the highest descendant critical point of x; if $x \in C_1[\mathcal{P}]$, then $v_x = x$. Moreover, let $w_x \in C_2[\mathcal{P}]$ be the highest descendant critical point of $\alpha(v_x)$. Let $h_x \coloneqq \max(f_1(x) + \delta', f_2(w_x))$. We define $\alpha'(x)$ as the ancestor of w_x at height h_x, that is, $\alpha'(x) \coloneqq {w_x}|^{h_x}$.

Properties of α'. For each point $x \in T_1$, we have the following three properties:

(i) the height of $\alpha'(x)$ is at least $f_1(x) + \delta'$,

(ii) $\alpha'(x)$ is an ancestor of w_x,

(iii) $\alpha'(x)$ is a descendant of $\alpha(x)$.

To see that property **(iii)** holds, observe that as x is an ancestor of v_x, the point $\alpha(x)$ must be ancestor of $\alpha(v_x)$, and thus also of w_x. If $h_x = f_2(w_x)$ then $\alpha'(x) = w_x$ and the property holds. Otherwise, if $h_x = f_1(x) + \delta'$, then because the height of $\alpha(x)$ is at least $f_1(x) + \delta'$ the property also holds. We show that α' is an up-map that extends φ.

α' is an up-map. By property **(i)** we immediately know that the height of the image of each point x is at least the height of x. To show that α' preserves ancestors, let $x_1, x_2 \in T_1$ such that $x_1 \preceq x_2$. If $v_{x_1} = v_{x_2}$, then it follows that $w_{x_1} = w_{x_2}$. Property **(ii)** tells us that both $\alpha'(x_1)$ and $\alpha'(x_2)$ are ancestors of w_{x_2}, so as $f_1(x_1) \leq f_1(x_2)$ we infer that $\alpha'(x_1) \preceq \alpha'(x_2)$. Otherwise, if $v_{x_1} \neq v_{x_2}$, then we must have $v_{x_1} \preceq x_1 \preceq v_{x_2}$. As α preserves ancestors, we know that $\alpha(v_{x_1})$ is a descendant of $\alpha(v_{x_2})$, and thus also that w_{x_1} is a descendant of w_{x_2}. Again, both $\alpha'(x_1)$ and $\alpha'(x_2)$ are ancestors of w_{x_1}, so $\alpha'(x_1) \preceq \alpha'(x_2)$. Next, to see that α' extends φ, let $x \in \mathrm{dom}(\varphi)$. The point x is a critical point of T_1, so $v_x = x$. Similarly, the point $\varphi(x)$ is a critical point of T_2. Since α extends φ, the point $\alpha(x)$ is an ancestor of $\varphi(x)$, so w_x is either equal to or an ancestor of $\varphi(x)$. Either way, by property **(ii)** it follows that $\alpha'(x)$ is an ancestor of $\varphi(x)$. Finally, we show that the φ-shift of α' is δ'. Consider an arrow $(x, \alpha'(x))$ whose shift is strictly greater than δ'. We argue that $(x, \alpha'(x))$ is in the fan of φ; this directly implies that the φ-shift of α' is δ'. By construction, it must be the case that $h_x = f_2(w_x) > f_1(x) + \delta'$. In other words, it holds that $\sigma(x, w_x) > \delta'$. The point v_x is a descendant of x and thus $f_1(v_x) \leq f_1(x)$. Therefore, we obtain $\sigma(v_x, w_x) \geq \sigma(x, w_x) > \delta'$. As both v_x and w_x are critical points, the value $\delta^* \coloneqq \sigma(v_x, w_x)$ is a critical value. By choice of δ' we thus know that $\delta^* \geq \delta$. We claim that $\sigma(v_x, \alpha(v_x)) > \delta$. If $\delta^* = \delta$, then because we assumed that $\mathcal{I}$ does not use a critical pair that corresponds to δ, we must have $\alpha(v_x) \succ w_x$. The claim follows. Otherwise, if $\delta^* > \delta$, then since $\alpha(v_x) \succeq w_x$ we immediately get $\sigma(v_x, \alpha(v_x)) > \delta$. So our claim holds. Since the φ-shift of α is at most δ, we know that $\sigma_{\mathcal{P}}(v_x, \alpha(v_x)) = 0$. This means that the arrow $(v_x, \alpha(v_x))$ is in the fan of φ. In other words, there is a descendant v' of v_x such that $v' \in \mathrm{dom}(\varphi)$ and $\varphi(v') = w_x$. As a result, the point $\alpha(v_x)$ is a critical point of T_2, so $\alpha(v_x) = w_x$. Moreover, since $h_x = f_2(w_x)$, we

know that $\alpha'(x) = w_x$. Hence, the pair $(x, \alpha'(x))$ is also in the fan of φ. So, α' has φ-shift δ.

$\mathcal{I}'$ is an interleaving. Using analogous arguments, the map β' is an up-map that extends ψ and has ψ-shift δ. It remains to show that (α', β') is an interleaving, that is, $\beta'(\alpha'(x)) \succeq x$ for all $x \in T_1$, and $\alpha'(\beta'(y)) \succeq y$ for all $y \in T_2$. We prove that this statement holds for all $x \in T_1$; an analogous argument shows the symmetric case. We first show that this holds for all vertices of T_1. Let $v \in V(T_1)$. From property **(i)** we know that the height of $\beta'(\alpha'(v))$ is at least $f_1(v) + 2\delta'$. Moreover, because $\mathcal{I}$ is an interleaving, we know that $\beta(\alpha(v)) \succeq v$ holds. Therefore, it suffices to show that $\beta'(\alpha'(v))$ and $\beta(\alpha(v))$ lie on the same edge of T_1. In the remainder, we use $y \coloneqq \alpha(v)$ and $y' \coloneqq \alpha'(v)$.

Case $\sigma(v, y) > \delta$. The φ-shift of each arrow of α is at most δ. This means that if $\sigma(v, y) > \delta$, then the arrow (v, y) must lie in the fan of φ. In particular, this means that there is a point $u \in \mathrm{dom}(\varphi)$ such that $\varphi(u) = y$. So, the point y is a critical point of T_2. The vertex v itself is also a critical point, so it follows that $w_v = y$ and $h_v = f_2(y)$. By construction of α', we thus have $y' = y$. Similarly, the highest descendant critical point of y is the point y itself. Let $v_y \in C_1[\mathcal{P}]$ be the highest descendant critical point of $\beta(y)$. Since all vertices of T_1 are critical points, the point $\beta(y)$ must lie on the same edge as v_y. By construction of β', we know that $\beta'(y)$ is an ancestor of v_y. Combining with property **(iii)**, it follows that the points $\beta'(y') = \beta'(y)$ and $\beta(y)$ lie on the same edge.

Case $\sigma(v, y) \leq \delta$. Recall that w_v is the highest descendant critical point of y. Let $w_{y'}$ denote the highest descendant critical point of y'. We claim that $w_{y'} = w_v$. From property **(ii)**, it follows that $w_{y'}$ is an ancestor of w_v. On the other hand, by property **(iii)** we know that y' is a descendant of y. As a result, the point y' either is a descendant of w_v or it lies on an edge between w_v and y. In both cases, we must have $w_{y'} \preceq w_v$. Combining, we thus obtain $w_v \preceq w_{y'} \preceq w_v$, which means our claim holds. We distinguish two more cases.

- First consider the case that $\sigma(y, \beta(y)) > \delta$. The ψ-shift of β is

at most δ, so the arrow $(y, \beta(y))$ must lie in the fan of ψ. In particular, this means that there is a point $y^* \in \mathrm{dom}(\psi)$ that is a descendant of y and satisfies $\psi(y^*) = \beta(y)$. Moreover, we know that the point $\beta(y)$ is a critical point. The point y^* is a critical point of T_2, so y^* must be a descendant of w_v. Since $w_{y'} = w_v$, it follows that y^* is also a descendant of $w_{y'}$. From property **(ii)** it then follows that $\beta'(y')$ is an ancestor of $\beta(y)$. Putting everything together, we obtain $\beta'(y') \succeq \beta(y) \succeq v$.

- Lastly, we consider the case that $\sigma(y, \beta(y)) \leq \delta$. Let $v^* \in V(T_1)$ be the lower endpoint of the edge that contains $\beta(y)$. Both v and v^* are vertices, so the value $\delta^* \coloneqq \frac{1}{2}(f_1(v^*) - f_1(v))$ is a critical value. Since $\beta(y) \succeq v^*$, we obtain $\delta^* \leq \frac{1}{2}(\sigma(v, y) + \sigma(y, \beta(y)))$. In our current case, we assumed both summands to be at most δ, so it follows that $\delta^* \leq \delta$. We claim that $\delta^* < \delta$. Otherwise, if $\delta^* = \delta$, then both $\sigma(v, y)$ and $\sigma(y, \beta(y))$ must be equal to δ, and we get $\beta(y) = v^*$. However, we assumed that $\mathcal{I}$ does not use any critical pairs that correspond to δ. Hence, it holds that $\delta^* < \delta$. By choice of δ', we thus have $\delta^* \leq \delta'$. By property **(iii)** for the map β', we get $\beta'(y') \preceq \beta(y')$. Similarly, by property **(iii)** for the map α', we get $y' \preceq y$. Since β preserves ancestors, it follows that $\beta(y') \preceq \beta(y)$. Combining, we obtain $\beta'(y') \preceq \beta(y') \preceq \beta(y)$. Lastly, by property **(i)**, we know that the height of $\beta'(y')$ is at least $f_1(v) + 2\delta'$. So, the point $\beta'(y')$ lies between the points v^* and $\beta(y)$. In particular, this means that $\beta(y)$ and $\beta'(y')$ lie on the same edge. We conclude that $\beta'(y')$ must be an ancestor of v.

This shows that $\beta'(\alpha'(v))$ is an ancestor of v. To conclude the proof, consider a point x of T_1 and let $v_x \in V(T_1)$ be the lower endpoint of the edge that contains x. Since α' preserves ancestors, we obtain $\alpha'(x) \succeq \alpha'(v_x)$. Similarly, since β' preserves ancestors, we get $\beta'(\alpha'(x)) \succeq \beta'(\alpha'(v_x))$. As v_x is a vertex, we know that $\beta'(\alpha'(v_x)) \succeq v_x$. Combining, we get $\beta'(\alpha'(x)) \succeq v_x$. Lastly, we know that $x \succeq v_x$, so it follows that $\beta'(\alpha'(x)) \succeq x$. □

It directly follows that the $\mathcal{P}$-residual interleaving distance uses a realizing critical pair and that the $\mathcal{P}$-residual interleaving distance attains a $\mathcal{P}$-critical value.

Corollary 13. If $\mathcal{P}$ is finite, then any optimal $\mathcal{P}$-extension uses a realizing critical pair.

Corollary 14. If $\mathcal{P}$ is finite, then $d_{\mathcal{P}}(T_1, T_2) \in \Delta[\mathcal{P}](T_1, T_2)$.

Let $\delta_1 \in \Delta[\mathcal{P}]$ be the greatest value that satisfies $\delta_1 \leq d_{\mathcal{P}}$ and let $\delta_2 \in \Delta[\mathcal{P}]$ be the smallest value that satisfies $\delta_2 > d_{\mathcal{P}}$. By definition of $d_{\mathcal{P}}$, for all $\varepsilon > 0$ there is a $\mathcal{P}$-extension with $\mathcal{P}$-shift $d_{\mathcal{P}} + \varepsilon$. In particular, for ε small enough, there is a $\mathcal{P}$-extension $\mathcal{I}$ with $\mathcal{P}$-shift $\delta_1 < \sigma_{\mathcal{P}}(\mathcal{I}) < \delta_2$. In other words, there are no critical pairs that correspond to $\sigma_{\mathcal{P}}(\mathcal{I})$. This means that $\mathcal{I}$ cannot use a critical pair that corresponds to the $\mathcal{P}$-shift of $\mathcal{I}$. By Lemma 12, it then follows that there exists a $\mathcal{P}$-extension with $\mathcal{P}$-shift $\delta_1 = d_{\mathcal{P}}$.

Theorem 15. Any finite partial interleaving $\mathcal{P}$ between any two merge trees T_1 and T_2 admits an optimal $\mathcal{P}$-extension.

Isolated extensions. We conclude this section by arguing that there always exist "isolated" optimal extensions, that is, extensions that satisfy two finiteness properties. Specifically, this is useful when reasoning about "greatest" shifts and "highest" ancestor. Formally, assume the $\mathcal{P}$-residual interleaving distance is strictly positive, that is, $d_{\mathcal{P}} > 0$. We say an optimal $\mathcal{P}$-extension $\mathcal{I}$ is isolated if (1) the $\mathcal{P}$-shift of only finitely many arrows of $\mathcal{I}$ is equal to $d_{\mathcal{P}}$, and (2) there exists some $\varepsilon > 0$ such that for each arrow (x, y) of $\mathcal{I}$ with $\mathcal{P}$-shift at least $d_{\mathcal{P}} - \varepsilon$, there exists a descendant x' of x, such that (x', y) is an arrow in $\mathcal{I}$ with the same target and $\mathcal{P}$-shift $d_{\mathcal{P}}$. We show that there always exists an isolated $\mathcal{P}$-extension.

Lemma 16. If $\mathcal{P}$ is finite and $d_{\mathcal{P}} > 0$, there exists an isolated $\mathcal{P}$-extension.

Proof. By Theorem 15, there exists an optimal $\mathcal{P}$-extension $\mathcal{I} = (\alpha, \beta)$. We use $\mathcal{I}$ to construct an isolated $\mathcal{P}$-extension $\mathcal{I}' = (\alpha', \beta')$. We describe the construction of the map α' from T_1 to T_2; the construction of the map β' from T_2 to T_1 is symmetric. Let $X \coloneqq C_1[\mathcal{P}] \cup \beta(C_2[\mathcal{P}])$ and $Y \coloneqq C_2[\mathcal{P}] \cup \alpha(C_1[\mathcal{P}])$. There are only finitely many critical points of both T_1 and T_2, so the sets X and Y are finite. Since every leaf is critical, every point x of T_1 has a descendant in X, and by finiteness a highest descendant in X, which we denote v_x. Let $h_x \coloneqq \max(f_1(x), f_2(\alpha(v_x)))$. We define $\alpha'(x) = \alpha(v_x)|^{h_x}$.

α' is an up-map with φ-shift at most $d_{\mathcal{P}}$. Let $x \in T_1$. We have $h_x \geq f_1(x)$, so by construction $f_2(\alpha'(x)) \geq f_1(x)$. To see that α' preserves ancestors, let x_1 and x_2 be points of T_1 and suppose x_2 is an ancestor of x_1. Then v_{x_2} is an ancestor of v_{x_1}. By construction, it then follows that $\alpha'(x_2)$ is an ancestor of $\alpha(v_{x_2})$, and hence also of $\alpha(v_{x_1})$. Moreover, since $f_1(x_2) \geq f_1(x_1)$, it follows that $h_{x_2} \geq h_{x_1}$. We conclude that $\alpha'(x_2)$ must be an ancestor of $\alpha'(x_1)$. So α' is an up-map. Next, we argue that α' extends φ. Let $x \in \mathrm{dom}(\varphi)$. Then x is a critical point of T_1, so $x \in X$ and $v_x = x$. By construction, $\alpha'(v_x) = \alpha(v_x)$, and since α extends φ it follows that $\alpha'(x)$ is ancestor of $\varphi(x)$. So indeed, α' extends φ. Lastly, to prove that the φ-shift of α' is at most $d_{\mathcal{P}}$, it suffices to show that for all points x it holds that $\alpha'(x)$ is a descendant of $\alpha(x)$. If $h_x = f_1(x)$ this immediately follows. Otherwise, $h_x = f_2(\alpha(v_x))$, and we know that $\alpha'(x) = \alpha(v_x)$. The map α preserves ancestors; since v_x is a descendant of x, it follows that the point $\alpha'(x)$ is a descendant of $\alpha(x)$. Since $\mathcal{I}$ is an optimal interleaving, we conclude that the $\mathcal{P}$-shift of α' is bounded by $d_{\mathcal{P}}$.

$\mathcal{I}'$ is an interleaving. For any vertex v, its highest descendant in the set X, is v itself. So, by construction, we have $\alpha'(v) = \alpha(v)$. Moreover, as the vertex v is a critical point, $\alpha(v)$ is in the set Y. So, by construction of β', we have $\beta'(\alpha'(v)) = \beta(\alpha(v))$. Therefore, because $\mathcal{I}$ is an interleaving, $\beta'(\alpha'(v))$ is an ancestor of v. Now, for an arbitrary point $x \in T_1$, let $v \in V(T_1)$ be the lower endpoint of the edge that contains x. As described above, the point $\beta'(\alpha'(v))$ is an ancestor of v. Moreover, since α' preserves ancestors, $\alpha'(x)$ is an ancestor of $\alpha'(v)$. Similarly, since β' preserves ancestors, $\beta'(\alpha'(x))$ is an ancestor of $\beta'(\alpha'(v))$. Combining, we get that $\beta'(\alpha'(x))$ is an ancestor of v. As x is an ancestor of v, and $f_1(\beta'(\alpha'(x)) \geq f_1(x)$, we obtain that $\beta'(\alpha'(x))$ is an ancestor of x. Therefore $\mathcal{I}'$ is an interleaving.

$\mathcal{I}'$ is isolated. To see that property (1) holds, we want to bound the number of arrows with $\mathcal{P}$-shift $d_{\mathcal{P}}$. Let $x \in T_1$. By construction, the point $\alpha'(x)$ satisfies either (a) $f_2(\alpha'(x)) = f_1(x)$ or (b) $\alpha'(x) = \alpha(v_x)$. In case (a), the $\mathcal{P}$-shift of the arrow $(x, \alpha'(x))$ is 0, so it suffices to bound case (b). By definition, v_x is a critical point. Since $\mathcal{P}$ is finite, there are only finitely many critical points. Hence, there are only finitely many points y of T_2 that can satisfy $y = \alpha'(x)$. For

any such point y, each edge of T_1 contains at most one point for which the corresponding arrow has shift $d_{\mathcal{P}}$. Hence, there are only finitely many arrows of α' with $\mathcal{P}$-shift $d_{\mathcal{P}}$. Analogously, there are also only finitely many arrows of β' with $\mathcal{P}$-shift $d_{\mathcal{P}}$. Lastly, we show that property (2) holds. Let $\varepsilon > 0$ be such that there is no point $v \in X$ for which $d_{\mathcal{P}} - \varepsilon < f_2(\alpha(v)) - f_1(v) < d_{\mathcal{P}}$, and such that $d_{\mathcal{P}} - \varepsilon > 0$. Since there are only finitely many points in X, such an ε exists. Consider an arrow a of $\mathcal{I}'$ with $\mathcal{P}$-shift at least $d_{\mathcal{P}} - \varepsilon$. Without loss of generality assume $a = (x, \alpha'(x))$ for some $x \in T_1$. Since the $\mathcal{P}$-shift of a is positive, we have $h_x = f_2(\alpha(v_x))$. In other words, we have $\alpha'(x) = \alpha(v_x)$. As v_x is a descendant of x, it holds that $f_2(\alpha'(x)) - f_1(v_x) > f_2(\alpha'(x)) - f_1(x)$. Moreover, by construction, we have $\alpha'(v_x) = \alpha(v_x)$, and thus $\alpha'(v_x) = \alpha'(x)$. As $v_x \in X$, by our choice of ε, the shift of $(v_x, \alpha'(v_x))$ is equal to $d_{\mathcal{P}}$. We conclude that property (2) holds. □

4 Locally Correct Interleavings

The interleaving distance is a bottleneck distance: it yields a single value that quantifies the worst discrepancy between the two merge trees. While this global value is useful for comparison, we are interested in the actual interleavings that realize the distance. Importantly, we seek interleavings that are not only globally optimal, but also locally meaningful. Specifically, if a part of the interleaving is fixed, we want the remainder to be optimal relative to this fixed part. To formalize this notion of local optimality, we build upon the residual interleaving distance introduced in the previous section. Let (T_1, f_1) and (T_2, f_2) be two merge trees, and fix a partial interleaving $\mathcal{P} = (\varphi, \psi)$. For a subset $S_1 \subseteq \mathrm{dom}(\varphi)$, we define the restriction of φ to S_1 as the map $\varphi|_{S_1} \colon S_1 \to T_2$ given by $\varphi|_{S_1}(x) \coloneqq \varphi(x)$ for all points x of S_1. Any restriction of φ is a partial up-map, and φ extends all of its restrictions. The restriction of ψ to a subset $S_2 \subseteq T_2$ is defined symmetrically. Lastly, the pair $\mathcal{R} = (\varphi|_{S_1}, \psi|_{S_2})$ is a partial interleaving such that $\mathcal{P}$ extends $\mathcal{R}$. We call $\mathcal{R}$ a restriction of $\mathcal{P}$. A complete interleaving $\mathcal{I}$ is locally correct if for all restrictions $\mathcal{R}$ of $\mathcal{I}$, the $\mathcal{R}$-residual shift of $\mathcal{I}$ is equal to the $\mathcal{R}$-residual interleaving distance.

Definition 17. A complete interleaving $\mathcal{I} = (\alpha, \beta)$ is locally correct if and only if for all $S_1 \subseteq T_1$ and $S_2 \subseteq T_2$, the restriction $\mathcal{R} =$

$(\alpha|_{S_1}, \beta|_{S_2})$ satisfies

$$d_{\mathcal{R}}(T_1, T_2) = \sigma_{\mathcal{R}}(\mathcal{I}). \tag{3}$$

See Figure 10 for some examples of interleavings that are (not) locally correct. Let $\mathcal{I}$ be an arbitrary, not necessarily locally correct, interleaving. For any restriction $\mathcal{R}$ of $\mathcal{I}$, we know that $\mathcal{I}$ extends $\mathcal{R}$. Hence, the $\mathcal{R}$-residual interleaving distance is at most the $\mathcal{R}$-shift of $\mathcal{I}$. To see that an interleaving $\mathcal{I}$ is locally correct, it hence suffices to show that for all restrictions $\mathcal{R}$ of $\mathcal{I}$ it holds that $d_{\mathcal{R}}(T_1, T_2) \geq \sigma_{\mathcal{R}}(\mathcal{I})$.

Observation 18. For any restriction $\mathcal{R}$ of $\mathcal{I}$, it holds that $d_{\mathcal{R}}(T_1, T_2) \leq \sigma_{\mathcal{R}}(\mathcal{I})$.

Suppose $\mathcal{I}$ is locally correct. Taking $\mathcal{R}$ as the empty restriction, we see that $\mathcal{I}$ satisfies $d(T_1, T_2) = \sigma(\mathcal{I})$. In other words, any locally correct interleaving is an optimal interleaving. The reverse is not necessarily true; not every optimal interleaving is locally correct (see Figure 2). That raises the question: does a locally correct interleaving always exist? In the remainder of this section, we answer this question affirmatively.

Theorem 19. Any two merge trees T_1 and T_2 admit a locally correct interleaving.

Overview of the proof. To prove Theorem 19, we give an explicit construction of a locally correct interleaving. Specifically, we incrementally build a partial interleaving $\mathcal{P}$ by augmenting it

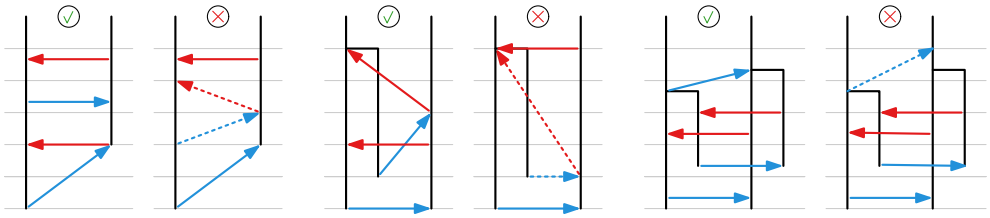


Figure 10: Three instances of T_1 and T_2 that together showcase all the types of critical pairs. For each instance, we show an interleaving that is locally correct (✓), and one that is not (✗). The non-dashed arrows in the ✗ cases form restrictions for which Equation (3) does not hold.

with "bottleneck" arrows. We show that with every augmentation, the $\mathcal{P}$-residual interleaving distance strictly decreases, until, after a finite number of iterations, it becomes zero. To conclude the proof, we then show that any complete interleaving that extends the resulting partial interleaving $\mathcal{P}$ is locally correct.

4.1 Bottlenecks

The interleaving distance is determined by the greatest shift among all arrows of an optimal interleaving. Locally, we might improve such an interleaving by "pushing down" some of the arrows, thereby decreasing their shifts. However, we cannot do this for all arrows; otherwise, we would obtain an interleaving whose shift is strictly smaller than the interleaving distance. We are interested in a bottleneck: a set of arrows that we cannot "push down". In this section, we give a characterization of such a bottleneck, and we argue that, under some mild assumptions, all arrows in a bottleneck correspond to realizing critical pairs. We first define the relative difference of two, not necessarily related, partial up-maps φ and φ'. Specifically, consider the set of points for which either φ is defined and φ' is not, or for which both are defined but where they do not agree:

$$S \coloneqq \mathrm{dom}(\varphi) \setminus \{x \in \mathrm{dom}(\varphi) \cap \mathrm{dom}(\varphi') : \varphi(x) = \varphi'(x)\}.$$

We define the relative difference of φ and φ' as the restriction of φ to S, denoted $\varphi \setminus \varphi' \coloneqq \varphi|_S$. Similarly, for two partial interleavings $\mathcal{P} = (\varphi, \psi)$ and $\mathcal{P}' = (\varphi', \psi')$, we define the relative difference $\mathcal{P} \setminus \mathcal{P}'$ as the pair of relative differences $(\varphi \setminus \varphi', \psi \setminus \psi')$. The resulting pair is a restriction of $\mathcal{P}$, so it directly follows that any relative difference is a partial interleaving.

Augmentations. Fix a partial interleaving $\mathcal{P} = (\varphi, \psi)$. In the remainder of this section, we assume that $\mathcal{P}$ is finite, and that the $\mathcal{P}$-residual interleaving distance is strictly positive, that is, $d_{\mathcal{P}} > 0$. A $\mathcal{P}$-extension $\mathcal{Q}$ is a $\mathcal{P}$-augmentation if (i) the $\mathcal{P}$-residual shift of $\mathcal{Q}$ is at most $d_{\mathcal{P}}$, that is, $\sigma_{\mathcal{P}}(\mathcal{Q}) \leq d_{\mathcal{P}}$, and (ii) the $\mathcal{Q}$-residual interleaving distance is strictly less than $d_{\mathcal{P}}$, that is, $d_{\mathcal{Q}} < d_{\mathcal{P}}$. We say that $\mathcal{Q}$ is minimal if it does not extend any other $\mathcal{P}$-augmentation.

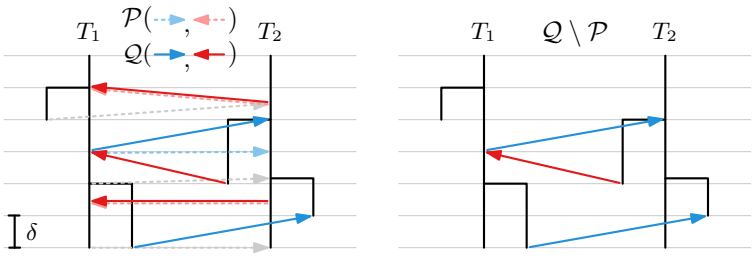


Figure 11: A partial interleaving $\mathcal{P}$ with $d_{\mathcal{P}} = \delta$, and a minimal augmentation $\mathcal{Q}$ with $d_{\mathcal{Q}} < \delta$; the dashed gray arrows showcase an optimal $\mathcal{Q}$-extension (left). The bottleneck $\mathcal{Q} \setminus \mathcal{P}$ (right).

Lemma 20. Any finite interleaving $\mathcal{P}$ with $d_{\mathcal{P}} > 0$ admits a finite minimal $\mathcal{P}$-augmentation.

Proof. By Lemma 16, there exists an isolated $\mathcal{P}$-extension. Let $\mathcal{I}$ be such an extension. Then in $\mathcal{I}$, only finitely many arrows of $\mathcal{P}$ have $\mathcal{P}$-shift $d_{\mathcal{P}}$. Let $\mathcal{Q}$ be the partial interleaving that consists of the arrows of $\mathcal{P}$ and the arrows of $\mathcal{I}$ with $\mathcal{P}$-shift $d_{\mathcal{P}}$. If a point x of T_1 occurs in two arrows (x, y_1) and (x, y_2), with $f_2(y_1) \leq f_2(y_2)$, we take the arrow (x, y_2). We choose arrows symmetrically if a point of T_2 occurs in two arrows. Since $\mathcal{I}$ extends $\mathcal{P}$, it directly follows that $\mathcal{Q}$ extends $\mathcal{P}$ as well. We now argue that $\mathcal{Q}$ is an augmentation. By construction, each arrow of $\mathcal{Q}$ has $\mathcal{P}$-shift at most $d_{\mathcal{P}}$, so property (i) holds. To see that property (ii) holds; observe that $\mathcal{I}$ is a $\mathcal{Q}$-extension. Since $\mathcal{I}$ is isolated, there is an $\varepsilon > 0$ such that for each arrow (x, y) of $\mathcal{I}$ with $\mathcal{P}$-shift at least $d_{\mathcal{P}} - \varepsilon$, there exists a descendant x' of x such that (x', y) is an arrow of $\mathcal{I}$. By construction, the arrow (x', y) is an arrow of $\mathcal{Q}$. In other words, the $\mathcal{Q}$-shift of the arrow (x, y) is 0. Additionally, the $\mathcal{Q}$-shift of each arrow of $\mathcal{P}$ is at most $d_{\mathcal{P}}$. Together, it follows that all arrows of $\mathcal{I}$ with shift at least $d_{\mathcal{P}} - \varepsilon$ have $\mathcal{Q}$-shift 0. Hence, the $\mathcal{Q}$-residual interleaving distance is at most $d_{\mathcal{P}} - \varepsilon < d_{\mathcal{P}}$. We conclude that $\mathcal{Q}$ is an augmentation with a finite number of arrows, which implies that there exists a finite minimal augmentation. □

Let $\mathcal{Q}$ be a minimal $\mathcal{P}$-augmentation. We refer to the relative difference $\mathcal{Q} \setminus \mathcal{P}$ as a $\mathcal{P}$-bottleneck. Intuitively, a bottleneck captures a minimal set of arrows that we need to "add" to decrease the residual interleaving distance. See Figure 11 for an illustra-

tion. Recall (the proof of) Lemma 12. If we apply the lemma to a given $\mathcal{P}$-extension $\mathcal{I}$ that does not use a realizing critical pair, we can construct another $\mathcal{P}$-extension $\mathcal{I}'$ that has strictly smaller shift. The exact same proof still works for a slightly stronger statement: it suffices to assume that the relative difference $\mathcal{I} \setminus \mathcal{P}$ does not use a realizing critical pair.

Observation 21. Let $\mathcal{P}$ be a finite interleaving and let $\mathcal{I}$ be a $\mathcal{P}$-extension. If $\mathcal{I} \setminus \mathcal{P}$ does not use a realizing critical pair, there exists a $\mathcal{P}$-extension $\mathcal{I}'$ with $\sigma_{\mathcal{P}}(\mathcal{I}') < \sigma_{\mathcal{P}}(\mathcal{I})$.

We use this observation to prove the following lemma.

Lemma 22. If $\mathcal{P}$ is finite and $d_{\mathcal{P}} > 0$, any $\mathcal{P}$-bottleneck $\mathcal{Q} \setminus \mathcal{P}$ uses a realizing critical pair.

Proof. Let $\mathcal{I}$ be an optimal $\mathcal{Q}$-extension. Since $\mathcal{Q}$ extends $\mathcal{P}$, so does $\mathcal{I}$. Moreover, as $\mathcal{I}$ is optimal, the $\mathcal{Q}$-shift of $\mathcal{I}$ is equal to the $\mathcal{Q}$-residual interleaving distance, that is, $\sigma_{\mathcal{Q}}(\mathcal{I}) = d_{\mathcal{Q}}$. Since $\mathcal{Q}$ is an augmentation, the shift of all arrows of $\mathcal{Q} \setminus \mathcal{P}$ is at most $d_{\mathcal{P}}$, and $d_{\mathcal{Q}} < d_{\mathcal{P}}$. Combining, we obtain $\sigma_{\mathcal{P}}(\mathcal{I}) \leq d_{\mathcal{P}}$. In other words, $\mathcal{I}$ is an optimal $\mathcal{P}$-extension. By Observation 21, it follows that $\mathcal{I} \setminus \mathcal{P}$ uses a realizing critical pair b. As $d_{\mathcal{Q}} < d_{\mathcal{P}}$, we get that the $\mathcal{Q}$-shift of the arrows that correspond to b must be 0. In other words, $\mathcal{Q} \setminus \mathcal{P}$ uses b. □

A direct consequence of Lemma 22 is that if, for a given $\mathcal{P}$-extension $\mathcal{Q}$, the relative difference $\mathcal{Q} \setminus \mathcal{P}$ does not use a realizing critical pair, then $\mathcal{Q}$ cannot be a $\mathcal{P}$-augmentation. In other words, if the $\mathcal{P}$-shift of each arrow of $\mathcal{Q}$ is strictly less than the $\mathcal{P}$-residual interleaving distance, then the $\mathcal{Q}$-residual interleaving distance cannot be less than $d_{\mathcal{P}}$.

Corollary 23. Let $\mathcal{P}$ be a finite interleaving with $d_{\mathcal{P}} > 0$, and let $\mathcal{Q}$ be a $\mathcal{P}$-extension. If $\sigma_{\mathcal{P}}(a) < d_{\mathcal{P}}(T_1, T_2)$ for all arrows $a \in A_{\mathcal{Q}}$, then $d_{\mathcal{Q}}(T_1, T_2) \geq d_{\mathcal{P}}(T_1, T_2)$.

Dominant matchings. We say $\mathcal{P}$ is dominant if the shift of each arrow of $\mathcal{P}$ is strictly greater than $d_{\mathcal{P}}$. If $\mathcal{P}$ is dominant, we can characterize any minimal augmentation $\mathcal{Q}$. Recall that we say that $\mathcal{Q}$ uses $\mathcal{P}$ if they completely agree on the domains of $\mathcal{P}.\varphi$ and $\mathcal{P}.\psi$.

Lemma 24. Let $\mathcal{P}$ be a dominant, finite interleaving with $d_{\mathcal{P}} > 0$. For any minimal $\mathcal{P}$-augmentation $\mathcal{Q}$, (i) $\mathcal{Q}$ uses $\mathcal{P}$, and (ii) for each arrow a of $\mathcal{Q} \setminus \mathcal{P}$, it holds that $\sigma(a) = d_{\mathcal{P}}$.

Proof. We first show (i). Suppose for a contradiction that an arrow a of $\mathcal{P}$ is not used by $\mathcal{Q}$. Because $\mathcal{P}$ is dominant, the shift of a is at least $d_{\mathcal{P}}$. Moreover, since $\mathcal{Q}$ extends $\mathcal{P}$, there is an arrow a' that extends a. Therefore, the arrow a' is not in the fan of $\mathcal{P}$, so the $\mathcal{P}$-shift of a' is equal to the shift of a'. We obtain $\sigma_{\mathcal{P}}(\mathcal{Q}) \geq \sigma_{\mathcal{P}}(a') > \sigma(a) \geq d_{\mathcal{P}}$. However, $\mathcal{Q}$ is an augmentation: we have reached a contradiction. Property (i) follows. Next, we show (ii). By definition, the shift of a is at most $d_{\mathcal{P}}$, that is, $\sigma(a) \leq d_{\mathcal{P}}$. For a contradiction, assume that the shift of a is strictly less than $d_{\mathcal{P}}$, that is, $\sigma(a) < d_{\mathcal{P}}$. In particular, consider the partial interleaving $\mathcal{Q}'$ obtained after removing a from $\mathcal{Q}$. Since the shift of a is less than $d_{\mathcal{P}}$, it is not an arrow of $\mathcal{P}$. It follows that $\mathcal{Q}'$ extends $\mathcal{P}$. Lastly, any (complete) $\mathcal{Q}$-extension $\mathcal{I}$ is trivially also a $\mathcal{Q}'$-extension. Since $\sigma(a) < d_{\mathcal{P}}$, we then know that $\sigma_{\mathcal{Q}'}(\mathcal{I}) = \max(\sigma(a), \sigma_{\mathcal{Q}}(\mathcal{I})) \leq d_{\mathcal{P}}$. But then $\mathcal{Q}$ is not a minimal augmentation. □

Lemma 25. Let $\mathcal{P}$ be a dominant, finite interleaving with $d_{\mathcal{P}} > 0$. For any finite minimal $\mathcal{P}$-augmentation $\mathcal{Q}$, $\mathcal{Q} \setminus \mathcal{P}$ uses only realizing critical pairs.

Proof. For a contradiction, assume that $\mathcal{Q} \setminus \mathcal{P}$ uses an arrow $a = (x, y)$ that does not correspond to a critical pair. By Lemma 24, we know that the shift of a is exactly $d_{\mathcal{P}}$. Let $\mathcal{Q}'$ be the partial interleaving obtained by removing a from $\mathcal{Q}$. By construction, $\mathcal{Q}$ extends $\mathcal{Q}'$. We argue that $\mathcal{Q}'$ is an augmentation, thereby contradicting that $\mathcal{Q}$ is minimal. Without loss of generality, assume $x \in T_1$ and $y \in T_2$. Let $\mathcal{I} = (\alpha, \beta)$ be an optimal $\mathcal{Q}$-extension. As $\mathcal{Q}$ is a $\mathcal{P}$-augmentation, it follows that $\mathcal{I}$ is also an optimal $\mathcal{P}$-extension. If $\mathcal{I}$ uses an arrow (x', y) for some descendant $x' \preceq x$, then $\sigma(x', y) > \sigma(x, y) = d_{\mathcal{P}}$. Because $\mathcal{I}$ is an optimal $\mathcal{Q}$-extension, the $\mathcal{Q}$-shift of $\mathcal{I}$ is strictly less than $d_{\mathcal{P}}$, so the $\mathcal{Q}$-shift of (x', y) must be 0. In other words, the arrow (x', y) lies in the fan of $\mathcal{Q}.\varphi$, that is, there exists a descendant x'' of x' such that the arrow (x'', y) is used by $\mathcal{Q}$. By construction, the arrow (x'', y) is also used by $\mathcal{Q}'$. But then, the point x'' is a descendant of x, so the $\mathcal{Q}'$-shift of (x, y)

is also equal to 0. Hence, $\mathcal{I}$ is a $\mathcal{Q}'$-extension with $\mathcal{Q}'$-shift strictly smaller than $d_{\mathcal{P}}$. This implies that $\mathcal{Q}'$ is a $\mathcal{P}$-augmentation, and we reach a contradiction. For the remainder, we assume that for all strict descendants x' of x, the point $\alpha(x')$ is a strict descendant of y.

Claim 26. For some $\varepsilon > 0$, each strict descendant x' of x satisfies $f_2(\alpha(x')) < f_2(y) - \varepsilon$.

Proof. As $\mathcal{Q}$ is a $\mathcal{P}$-augmentation, it holds that $d_{\mathcal{Q}} < d_{\mathcal{P}}$. Moreover, as $\mathcal{Q}$ is finite, there are only finitely many $\mathcal{Q}$-critical values. By Corollary 14, both $d_{\mathcal{P}}$ and $d_{\mathcal{Q}}$ are equal to $\mathcal{Q}$-critical values. Hence, there exists an $\varepsilon' > 0$ such that $d_{\mathcal{Q}} + \varepsilon' < d_{\mathcal{P}}$. If there is no strict descendant x' of x with $f_2(\alpha(x')) \geq f_2(y) - \varepsilon'$, then our claim immediately holds if we take $\varepsilon = \varepsilon'$. Otherwise, let $y' = \alpha(x')$; we assumed that $y' \neq y$. It follows that the arrow (x', y') satisfies $\sigma(x', y') > d_{\mathcal{Q}}$. Since $\mathcal{I}$ is an optimal $\mathcal{Q}$-extension, we must have $\sigma_{\mathcal{Q}}(x', y') \leq d_{\mathcal{Q}}$; we obtain $\sigma_{\mathcal{Q}}(x', y') = 0$. So, the arrow (x', y') lies in the fan of $\mathcal{Q}.\varphi$. In other words, there is a descendant x'' of x' such that the arrow (x'', y') is used by $\mathcal{Q}$. Now, the points y' and y are two distinct $\mathcal{Q}$-critical points. There are only finitely many $\mathcal{Q}$-critical points, so there exists ε'' such that $f_2(y') + \varepsilon'' < f_2(y)$. ◁

A direct consequence of Claim 26 is that if x is not a leaf, the point y must have a descendant. If x is a leaf, then y cannot be a leaf, since otherwise the pair (x, y) is an arrow critical pair. Either way, y cannot be a leaf. We construct a pair $\mathcal{I}' = (\alpha', \beta')$ that does not use a. For $\varepsilon > 0$, we use X^{ε} to denote the set of ancestors x' of x with $f_1(x') - f_1(x) < \varepsilon$, and we use Y_{ε} to denote the set of descendants y' of y with $f_2(y) - f_2(y') < \varepsilon$.

Case 1: y is a critical point. If y is a vertex, then for any critical point v of T_1, the pair (v, y) corresponds to a critical pair. In particular, this means that x cannot be a critical point. By Claim 26, we know that there exists an $\varepsilon_1 > 0$ such that (i) all descendants x' of x satisfy $f_2(\alpha(x')) < f_2(y) - \varepsilon_1$. Since x is not a critical point, it cannot be a leaf. Hence, the point x has a strict descendant x', which means that there must be a strict descendant y^* of y at height $f_2(y^*) = f_1(y) - \varepsilon_1$ that satisfies $y^* \succeq \alpha(x')$ for all x'. Moreover, let $\varepsilon_2 > 0$ be such that (ii) there is no $\mathcal{Q}$-critical point in X^{ε_2}.

Lastly, let $\varepsilon_3 > 0$ such that (iii) $d_{\mathcal{Q}} + 2\varepsilon_3 < d_{\mathcal{P}}$. Set $\varepsilon \coloneqq \min(\varepsilon_1, \varepsilon_2, \varepsilon_3)$. For $x' \in T_1$ and $y' \in T_2$, we define the maps α' and β' as follows:

$$\alpha'(x') \coloneqq \begin{cases} y^*|^{\max(f_1(x'), f_2(y)-\varepsilon)}, & \text{if } x' \in X^\varepsilon, \\ \alpha(x') & \text{otherwise,} \end{cases} \qquad \beta'(y') \coloneqq \begin{cases} x|^{f_1(x)+\varepsilon}, & \text{if } \beta(y') \in X^\varepsilon, \\ \beta(y') & \text{otherwise.} \end{cases}$$

Since $f_2(y^*) = f_2(y) - \varepsilon_1 \leq f_2(y) - \varepsilon$ and $f_1(x) \leq f_1(x) + \varepsilon$, both α' and β' are well-defined. We first argue that α' is an up-map. By construction, we have $f_2(\alpha(x')) \geq f_1(x')$ for all x' of T_1. To see that α' preserves ancestors, let $x_1 \preceq x_2$ be two points of T_1. If both x_1 and x_2 lie within, or if both lie outside of X^ε, then we immediately obtain $\alpha'(x_1) \preceq \alpha'(x_2)$. If x_1 lies within and x_2 lies outside of X^ε, we get $\alpha'(x_1) \preceq \alpha(x_1)$ and $\alpha'(x_2) = \alpha(x_2)$. Since α preserves ancestors, we obtain $\alpha'(x_1) \preceq \alpha'(x_2)$. Otherwise, if x_1 lies outside of X^ε and x_2 lies within X^ε, then, by choice of y^*, we know that $\alpha'(x_1)$ is a (strict) descendant of y^*. It directly follows that $\alpha'(x_1) = \alpha(x_1) \preceq \alpha'(x_2)$. By property (ii), there are no $\mathcal{Q}$-critical points, and hence no $\mathcal{P}$-critical points within X^ε. So, α' trivially extends $\mathcal{P}.\varphi$. By construction, the $\mathcal{Q}'.\varphi$-shift of α' is strictly less than $d_{\mathcal{P}}$. Next, we argue that β' is an up-map. Let $y' \in T_2$. Observe that, by construction, we have $\beta'(y')$ is an ancestor of $\beta(y')$. This directly implies that β' extends $\mathcal{P}.\psi$ and that $f_1(\beta'(y')) \geq f_2(y')$. To see that β' preserves ancestors, consider two points $y_1 \preceq y_2$ of T_2. The only non-trivial case is if $\beta(y_1) \in X^\varepsilon$ and $\beta(y_2) \notin X^\varepsilon$. Then, the point $\beta'(y_2) = \beta(y_2)$ must be ancestor of every point of X^ε. In particular, this means that $\beta'(y_2) \succeq x|^{f_1(x)+\varepsilon} = \beta'(y_1)$. So β' is an up-map. Lastly, we show that the $\mathcal{P}.\psi$-shift of β' is bounded by $d_{\mathcal{P}}$. Let $y' \in T_2$. If $\beta(y') \notin X$, then we directly obtain $\sigma_{\mathcal{P}.\psi}(y, \beta'(y)) = \sigma_{\mathcal{P}.\psi}(y, \beta(y)) \leq d_{\mathcal{P}}$. Otherwise, if $\beta(y') \in X^\varepsilon$, then we get $f_1(\beta'(y')) = f_1(x) + \varepsilon$. There is no critical point in X^ε, so we must have $\sigma(y', \beta(y')) < d_{\mathcal{Q}}$. So, by property (iii), it follows that $\sigma(y', \beta'(y')) \leq \sigma(y', \beta(y')) + \varepsilon < d_{\mathcal{Q}} + \varepsilon < d_{\mathcal{P}}$. As a result, the $\mathcal{Q}'.\psi$-shift of β' is strictly less than $d_{\mathcal{P}}$. Finally, we show that $\mathcal{I}'$ is an interleaving. Let $x' \in T_1$. If $x' \notin X^\varepsilon$, we get $\beta'(\alpha'(x')) = \beta'(\alpha(x')) \succeq \beta(\alpha(x')) \succeq x'$. Otherwise, if $x' \in X^\varepsilon$, then let v be the lower endpoint of the edge that contains x'. We know that v is not within X^ε, so $\beta'(\alpha'(v))$ is an ancestor of v. As α' is an up-map, we get $\alpha'(x')$ is an ancestor of $\alpha'(v)$. Similarly, since β' is an up-map, we get $\beta'(\alpha(x'))$ is an ancestor of $\beta'(\alpha'(v))$. So, $\beta'(\alpha'(x'))$ is an ancestor of v; it follows that it is also an ancestor of x'. Lastly, let $y' \in T_2$. If $\beta(y') \notin X^\varepsilon$, we get $\alpha'(\beta'(y')) = \alpha(\beta(y')) \succeq y'$. Other-

wise, if $\beta(y') \in X^\varepsilon$, we get $\beta'(y') = x|^{f_1(x)+\varepsilon} \succeq \beta(y')$. The point $\beta(y')$ is not within X^ε, so as α' preserves ancestors, $\alpha'(\beta(y'))$ is an ancestor of y'. Therefore $\alpha'(\beta'(y'))$ is an ancestor of y'. So, $\mathcal{I}'$ is an interleaving. Moreover, the $\mathcal{Q}'$-shift of $\mathcal{I}'$ is strictly less than $d_\mathcal{P}$.

Case 2: y is not a critical point. We choose ε as follows. First, let y^* be the highest critical descendant of y; this is a strict descendant of y. Let $\varepsilon_1 > 0$ be such that (i) $f_2(y^*) \leq f_2(y) - \varepsilon_1$. Secondly, let $\varepsilon_2 > 0$ be such that (ii) there are no $\mathcal{Q}$-critical points (beside potentially the point x itself) in the set X^{ε_2}. Such a value exists, because there are only finitely many $\mathcal{Q}$-critical points. Thirdly, let $\varepsilon_3 > 0$ be such that for all strict descendants x' of x, it holds that $f_2(\alpha(x')) < f_2(y) - \varepsilon_3$. Such a value ε_3 exists by Claim 26. Note that x does not necessarily have strict descendants; in that case any $\varepsilon_3 > 0$ suffices. Lastly, let $\varepsilon_4 > 0$ be such that $d_\mathcal{Q} + \varepsilon < d_\mathcal{P}$. Let $\varepsilon \coloneqq \min(\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4)$. For $x' \in T_1$ we define the map α' as follows:

$$\alpha'(x') \coloneqq \begin{cases} y^*|^{\max(f_1(x'), f_2(y)-\varepsilon)}, & \text{if } x' \in X^\varepsilon, \\ \alpha(x') & \text{otherwise.} \end{cases}$$

Observe that $f_2(y^*) \leq f_2(y) - \varepsilon_1 \leq f_2(y) - \varepsilon$. Hence, the map α' is well-defined. We now show that α' is an up-map that extends $\mathcal{P}.\varphi$. Let x' be a point of T_1. By construction, we directly have $f_1(\alpha'(x')) \geq f_1(x')$. To see that α' preserves ancestors, observe that $\alpha'(x')$ always lies on the same edge as $\alpha(x')$. Moreover, by choice of ε_3, the point y^* is an ancestor of all descendants of x. It follows that α' preserves ancestors. Next, to see that α' extends $\mathcal{P}.\varphi$, suppose $x' \in \mathrm{dom}(\mathcal{P}.\varphi)$. If $x' \notin X^\varepsilon$, then we have $\alpha'(x') = \alpha(x') \succeq \mathcal{P}.\varphi(x')$. Otherwise, by choice of ε, we must have $x' = x$. Since y^* is the highest descendant critical point of y, the point $\mathcal{P}.\varphi(x)$ must be a descendant of y^*. Hence, by construction, we indeed get $\alpha'(x) \succeq y^* \succeq \mathcal{P}.\varphi(x)$. Lastly, by construction, the $\mathcal{Q}'.\varphi$-shift of α' is strictly less than $d_\mathcal{P}$. To define the map β', we consider two more cases.

Case 2a: x is not a leaf. If x is not a leaf, we let $\beta' = \beta$. Trivially, β' is an up-map that extends $\mathcal{P}.\psi$ and whose $\mathcal{Q}'.\psi$-shift is strictly less than $d_\mathcal{P}$. It remains to show that the pair (α', β') is an interleaving. Let $x' \in T_1$. If $x' \notin X^\varepsilon$, we have $\beta'(\alpha'(x)) = \beta(\alpha(x))$, which

directly implies that $\beta'(\alpha'(x))$ is an ancestor of x. Otherwise, we have $x' \in X^\varepsilon$. Since x is not a leaf, there must be a descendant x'' of x. In particular, by choice of ε, it holds that $\alpha'(x'') = \alpha(x'')$ is a strict descendant of $\alpha'(x')$. The point x'' does not lie within X^ε, so we have $\beta'(\alpha'(x'')$ is an ancestor of x''. As β' preserves ancestors and $\alpha'(x')$ is an ancestor of $\alpha'(x'')$, we obtain that $\beta'(\alpha'(x'))$ is an ancestor of x''. It follows that $\beta'(\alpha'(x'))$ is an ancestor of x'. Lastly, let $y' \in T_2$; we have $\beta'(y') = \beta(y')$. Moreover, by construction, the points $\alpha'(\beta(y'))$ and $\alpha(\beta(y'))$ lie on the same edge. Since $\mathcal{I}$ is an interleaving, it directly follows that $\alpha'(\beta'(y'))$ is an ancestor of y'.

Case 2b: x is a leaf. If x is a leaf, we also modify β. Specifically, we use Y_ε to denote the set of descendants y' of y with height $f_2(y) - f_2(y') < \varepsilon$. For $y' \in T_2$, we define β' as follows:

$$\beta'(y') \coloneqq \begin{cases} \beta(y), & \text{if } y' \in Y_\varepsilon, \\ \beta(y') & \text{otherwise.} \end{cases}$$

We argue that β' is an up-map that extends $\mathcal{P}.\psi$. Let $y' \in T_2$. By construction, we get $f_1(\beta'(y')) \geq f_1(\beta(y')) \geq f_2(y')$. Moreover, if $y' \in Y_\varepsilon$, its image $\beta'(y')$ is equal to the image of its lowest ancestor not in Y_ε; it directly follows that β' preserves ancestors. Similarly, since $\beta'(y')$ is an ancestor of $\beta(y')$, it follows that if $y' \in \mathrm{dom}(\mathcal{P}.\psi)$ we have $\beta'(y')$ is an ancestor of $\mathcal{P}.\psi(y')$. So, β' is an up-map that extends $\mathcal{P}.\psi$. Next, we show that the $\mathcal{Q}'.\psi$-shift of β' is strictly less than by $d_\mathcal{P}$. Since $\mathcal{I}$ is an optimal $\mathcal{Q}$-extension, the $\mathcal{Q}$-shift of each arrow of $\mathcal{I}$ is bounded by $d_\mathcal{Q}$. Let $y' \in T_2$. By construction, $\mathcal{Q}'.\psi$ is equal to $\mathcal{Q}.\psi$. If y' does not lie within Y_ε, the $\mathcal{Q}'.\psi$-shift of $(y', \beta'(y'))$ is bounded by $d_\mathcal{Q}$, which is strictly less than $d_\mathcal{P}$. Otherwise, we have $y' \in Y_\varepsilon$. The $\mathcal{Q}$-shift of the arrow $(y, \beta(y))$ is bounded by $d_\mathcal{Q}$. This means that either (a) $\sigma(y, \beta(y)) \leq d_\mathcal{Q}$, or (b) the arrow $(y, \beta(y))$ lies inside the fan of $\mathcal{Q}.\psi$. In case (a), by choice of ε_4, it follows that $\sigma(y', \beta(y)) \leq d_\mathcal{Q} + \varepsilon < d_\mathcal{P}$. In case (b), there is a descendant y'' of y such that $y'' \in \mathrm{dom}(\mathcal{Q}.\psi)$ and $\mathcal{Q}.\psi(y'') = \beta(y)$. The point y'' is a $\mathcal{Q}$-critical point, so it must be a descendant of y^*. By choice of ε_1, it follows that y'' is a descendant of y'. Hence, the arrow $(y', \beta'(y')) = (y', \beta(y))$ lies within the fan of $\mathrm{dom}(\mathcal{Q}'.\varphi)$. In both cases, the $\mathcal{Q}'$-shift of $(y', \beta'(y'))$ is strictly less than $d_\mathcal{P}$. Hence, the $\mathcal{Q}'$-shift of β' is strictly less than $d_\mathcal{P}$. It remains to show that (α', β') is an interleaving. Observe that for all $y' \in T_2$,

the point $\beta'(y')$ is an ancestor of $\beta(y')$. Let $x' \in T_1$. If $\alpha'(x') = \alpha(x')$, then we obtain $\beta'(\alpha'(x')) = \beta'(\alpha(x')) \succeq \beta(\alpha(x'))$. As $\mathcal{I}$ is an interleaving, it follows that $\beta'(\alpha'(x'))$ is an ancestor of x'. Otherwise, if $\alpha'(x') \neq \alpha(x')$, then, by construction, the point $\alpha'(x')$ is an ancestor of y^*. In particular, this implies that $\beta'(\alpha'(x'))$ is an ancestor of $\beta(y)$. Since $\mathcal{I}$ is an optimal $\mathcal{Q}$-extension, we know that $\alpha(x) = y$. Moreover, it must be the case that $\beta(y)$ is an ancestor of x. It follows that $\beta'(\alpha'(x'))$ is also an ancestor of x, and thus also of x'. Finally, let $y' \in T_2$. Observe that for all $x' \in T_1$, the point $\alpha'(x')$ lies on the same edge as $\alpha(x')$. If $\beta'(y') = \beta'(y)$, then the point $\alpha'(\beta'(y')) = \alpha'(\beta(y'))$ lies on the same edge as $\alpha(\beta(y'))$. As $\mathcal{I}$ is an interleaving, it follows that $\alpha'(\beta'(y'))$ is an ancestor of y'. Otherwise, $\beta'(y')$ is an ancestor of $\beta(y')$. Since α' preserves ancestors, we get that $\alpha'(\beta'(y'))$ is an ancestor of $\alpha'(\beta(y'))$. The point $\alpha'(\beta(y'))$ lies on the same edge as $\alpha(\beta(y'))$, so again, as $\mathcal{I}$ is an interleaving, it follows that $\alpha'(\beta(y'))$ is an ancestor of y'. So, the point $\alpha'(\beta'(y'))$ is also an ancestor of y'.

Concluding the proof. In all cases, we obtain an interleaving $\mathcal{I}'$ that has $\mathcal{Q}'$-shift strictly less than $d_{\mathcal{P}}$. In other words, $\mathcal{Q}'$ is a $\mathcal{P}$-augmentation, and we reach a contradiction. □

4.2 Constructing a Locally Correct Interleaving

We show that a locally correct interleaving always exists by incrementally constructing one. Specifically, we iteratively replace $\mathcal{P}$ with a finite minimal $\mathcal{P}$-augmentation, and we show that after a finite number of iterations the $\mathcal{P}$-residual interleaving distance becomes zero, at which point we stop iterating. We conclude by showing that for the resulting partial interleaving $\mathcal{P}$, any optimal $\mathcal{P}$-extension is locally correct. For this, we maintain that $\mathcal{P}$ remains "partially" locally correct throughout the entire construction. Moreover, we show that after each step of the construction, $\mathcal{P}$ "specifies" at least one additional vertex of the input merge trees. Since each vertex can be specified at most once, it follows that the construction terminates.

Minimal augmentations. Formally, let $\mathcal{P}$ be a partial interleaving and assume $d_{\mathcal{P}} > 0$. Let $\mathcal{Q}$ be a finite minimal $\mathcal{P}$-

augmentation. By definition, $d_{\mathcal{Q}} < d_{\mathcal{P}}$. If for all restrictions $\mathcal{R}$ of $\mathcal{P}$ it holds that $d_{\mathcal{R}} \geq \sigma_{\mathcal{R}}(\mathcal{P})$, we say $\mathcal{P}$ is locally correct. Note that for a complete interleaving this definition is equivalent to Definition 17 (this follows from Observation 18).

Lemma 27. Let $\mathcal{P}$ be a finite interleaving and suppose $d_{\mathcal{P}} > 0$. Let $\mathcal{Q}$ be a finite minimal $\mathcal{P}$-augmentation. If $\mathcal{P}$ is dominant and locally correct, then so is $\mathcal{Q}$.

Proof. We first argue that $\mathcal{Q}$ is dominant. Let a be an arrow of $\mathcal{Q}$. If a is also an arrow of $\mathcal{P}$, then from the fact that $\mathcal{P}$ is dominant we know that $\sigma(a) > d_{\mathcal{P}} > d_{\mathcal{Q}}$. Otherwise, a must be an arrow of the bottleneck $\mathcal{Q} \setminus \mathcal{P}$. By Lemma 25, it then directly follows that $\sigma(a) = d_{\mathcal{P}} > d_{\mathcal{Q}}$. Next, we argue that $\mathcal{Q}$ is locally correct. Consider a restriction $\mathcal{R}$ of $\mathcal{Q}$; we need to show that $d_{\mathcal{R}} \geq \sigma_{\mathcal{R}}(\mathcal{Q})$. If $\mathcal{R} = \mathcal{Q}$, we immediately obtain $\sigma_{\mathcal{R}}(\mathcal{Q}) = 0 \leq d_{\mathcal{R}}$. Otherwise, let $\mathcal{R}'$ be the restriction of $\mathcal{R}$ to the domains of $\mathcal{P}.\varphi$ and $\mathcal{P}.\psi$. Since $\mathcal{Q}$ uses $\mathcal{P}$, it follows that $\mathcal{R}'$ uses $\mathcal{P}$ and is hence a restriction of $\mathcal{P}$. If $\mathcal{R}' = \mathcal{P}$, then by definition of an augmentation we know that the $\mathcal{R}'$-shift of any arrow of $\mathcal{Q}$ is $d_{\mathcal{P}}$. We obtain $\sigma_{\mathcal{R}'}(\mathcal{Q}) = d_{\mathcal{P}}$. Moreover, since $\mathcal{R} \neq \mathcal{Q}$, the $\mathcal{R}$-shift of at least one arrow of $\mathcal{Q} \setminus \mathcal{P}$ is non-zero. We obtain $\sigma_{\mathcal{R}}(\mathcal{Q}) = \sigma_{\mathcal{R}'}(\mathcal{Q}) = d_{\mathcal{P}}$. Since $\mathcal{Q}$ is a minimal augmentation and extends $\mathcal{R}$, we know that $\mathcal{R}$ cannot be an augmentation of $\mathcal{P}$. It follows that $d_{\mathcal{R}} \geq d_{\mathcal{P}}$. Putting it together, we obtain $d_{\mathcal{R}} = \sigma_{\mathcal{R}}(\mathcal{Q})$. Lastly, if $\mathcal{R}' \neq \mathcal{P}$, then since $\mathcal{P}$ is locally correct and $\mathcal{R}'$ is a restriction of $\mathcal{P}$, we have

$$d_{\mathcal{R}'} \geq \sigma_{\mathcal{R}'}(\mathcal{P}). \tag{4}$$

We first show that $\sigma_{\mathcal{R}'}(\mathcal{P}) \geq \sigma_{\mathcal{R}}(\mathcal{Q})$. First, none of the arrows of $\mathcal{R} \setminus \mathcal{R}'$ are arrows of $\mathcal{P}$, so $\sigma_{\mathcal{R}'}(\mathcal{P}) = \sigma_{\mathcal{R}}(\mathcal{P})$. Next, the $\mathcal{R}$-residual shift of $\mathcal{Q}$ is the maximum of the $\mathcal{R}$-residual shifts of $\mathcal{P}$ and $\mathcal{Q} \setminus \mathcal{P}$. Since $\mathcal{P}$ is dominant, we know that the shift of any arrow of $\mathcal{P}$ is strictly greater than $d_{\mathcal{P}}$. As $\mathcal{P} \setminus \mathcal{R}$ is not empty, it follows that $\sigma_{\mathcal{R}}(\mathcal{P}) > d_{\mathcal{P}}$. On the other hand, by definition of an augmentation, we know that the shift of any arrow of $\mathcal{Q} \setminus \mathcal{P}$ is at most $d_{\mathcal{P}}$. So, we get $\sigma_{\mathcal{R}}(\mathcal{Q}) = \max(\sigma_{\mathcal{R}}(\mathcal{P}), \sigma_{\mathcal{R}}(\mathcal{Q} \setminus \mathcal{P})) = \sigma_{\mathcal{R}}(\mathcal{P})$. Combining, we obtain $\sigma_{\mathcal{R}'}(\mathcal{P}) \geq \sigma_{\mathcal{R}}(\mathcal{Q})$. Next, we argue that $d_{\mathcal{R}} \geq d_{\mathcal{R}'}$. By construction, $\mathcal{R}$ extends $\mathcal{R}'$. As observed before, we have $\sigma_{\mathcal{R}'}(\mathcal{P}) = \sigma_{\mathcal{R}}(\mathcal{P}) > d_{\mathcal{P}}$. Combining with Equation (4), we obtain $d_{\mathcal{R}'} \geq \sigma_{\mathcal{R}'}(\mathcal{P}) > d_{\mathcal{P}}$. Let a be an arrow of $\mathcal{R} \setminus \mathcal{R}'$. Then a is arrow of the bottleneck $\mathcal{Q} \setminus \mathcal{P}$,

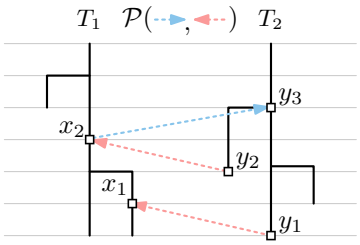


Figure 12: The points x_1, x_2, y_1, y_2 and y_3 are all critical, but only x_2, y_1 and y_2 are specified.

so the shift of a is exactly $d_{\mathcal{P}}$. In other words, $\sigma_{\mathcal{R}'}(a) = d_{\mathcal{P}} < d_{\mathcal{R}'}$. This means that we can apply Corollary 23, for the finite interleaving $\mathcal{R}'$ and the $\mathcal{R}'$-extension $\mathcal{R}$, to obtain $d_{\mathcal{R}} \geq d_{\mathcal{R}'}$. Thus, $d_{\mathcal{R}} \geq d_{\mathcal{R}'} \geq \sigma_{\mathcal{R}'}(\mathcal{P}) = \sigma_{\mathcal{R}}(\mathcal{Q})$, which concludes the proof. □

Recall that for each arrow (x, y), with $x \in T_1$ and $y \in T_2$, of $\mathcal{P}$, both x and y are critical points. However, only the point x has a "specified" target in the other tree. We say a point $x \in T_1$ or $y \in T_2$ is specified by $\mathcal{P}$, if $x \in \mathrm{dom}(\mathcal{P}.\varphi)$ or $y \in \mathrm{dom}(\mathcal{P}.\psi)$ respectively (see Figure 12). We argue that if $\mathcal{P}$ specifies all critical points that are not vertices, i.e., $C_1[\mathcal{P}] \setminus V(T_1) \subseteq \mathrm{dom}(\mathcal{P}.\varphi)$ and $C_2[\mathcal{P}] \setminus V(T_2) \subseteq \mathrm{dom}(\mathcal{P}.\psi)$, then so does any minimal augmentation.

Lemma 28. Let $\mathcal{P}$ be a finite interleaving and suppose $d_{\mathcal{P}} > 0$. Let $\mathcal{Q}$ be a finite minimal $\mathcal{P}$-augmentation. If $\mathcal{P}$ is dominant and specifies all critical points that are not vertices, then so does $\mathcal{Q}$. Moreover, $\mathcal{Q}$ specifies at least one vertex that $\mathcal{P}$ does not specify.

Proof. By Lemma 25, we know that the bottleneck $\mathcal{Q} \setminus \mathcal{P}$ uses only $\mathcal{P}$-critical pairs. Consider such a pair b. If b is an arrow critical pair, then without loss of generality assume $b = (x, y)$ for a point $x \in T_1$ and a point $y \in T_2$; the case $b = (y, x)$ is symmetric. By definition of an arrow critical pair, both x and y are $\mathcal{P}$-critical points. So the pair b does not create any additional $\mathcal{Q}$-critical points. As $\mathcal{P}$ already specifies all critical points that are not vertices and $\mathcal{Q}$ uses $\mathcal{P}$, we know that x must be a vertex. Since $\mathcal{P}$ is dominant, the shift of any arrow of $\mathcal{P}$ is strictly greater than $d_{\mathcal{P}}$. Moreover, the shift of any arrow of $\mathcal{Q} \setminus \mathcal{P}$ is exactly $d_{\mathcal{P}}$. Hence, the vertex x cannot be specified by $\mathcal{P}$: at least one new vertex is specified. If b is a zigzag critical pair, then without loss of generality

assume $b = (x_1, x_2)$ for two points x_1 and x_2 of T_1; if it is a pair of points of T_2 the argument is analogous. By definition of a zigzag critical pair, we know that both $x_1 \in V(T_1)$ and $x_2 \in V(T_1)$. Moreover, there is a point $y \in T_2$ such that $Q.\varphi(x_1) = y$ and $Q.\psi(y) = x_2$. So, x_1 is a vertex that is specified by $\mathcal{Q}$ but not by $\mathcal{P}$. Lastly, b creates a single $\mathcal{Q}$-critical point y, which is immediately specified. □

Finishing the proof of Theorem 19. Let $\mathcal{P}$ be the empty interleaving. Trivially, $\mathcal{P}$ is dominant and locally correct. Moreover, there are no (unspecified) critical points that are not vertices. Now, we iteratively replace $\mathcal{P}$ with a finite minimal $\mathcal{P}$-augmentation until the $\mathcal{P}$-residual interleaving distance becomes zero. By applying Lemmas 27 and 28 inductively, we maintain that $\mathcal{P}$ is dominant and locally correct, and specifies all critical points that are not vertices. Additionally, by Lemma 28, each iteration specifies at least one additional vertex. So after i iterations, at least i vertices are specified, and the total number of iterations is at most $|V(T_1)| + |V(T_2)|$. Afterwards, the residual interleaving distance is zero. Lemma 29 guarantees that a locally correct interleaving exists; its proof is similar to that of Lemma 27.

Lemma 29. Let $\mathcal{P}$ be a locally correct partial interleaving with $d_{\mathcal{P}} = 0$. Any optimal $\mathcal{P}$-extension is locally correct.

Proof. Fix a restriction $\mathcal{R}$ of $\mathcal{I}$. By Observation 18 it suffices to show that $d_{\mathcal{R}} \geq \sigma_{\mathcal{R}}(\mathcal{I})$. If $\mathcal{R}$ uses $\mathcal{P}$, then we immediately have $d_{\mathcal{R}} = 0 = \sigma_{\mathcal{R}}(\mathcal{I})$. Otherwise, let $\mathcal{R}'$ be the restriction of $\mathcal{R}$ to the domains of $\mathcal{P}.\varphi$ and $\mathcal{P}.\psi$. Observe that $\mathcal{R}'$ is also a restriction of $\mathcal{P}$. Since $\mathcal{P}$ is locally correct, we get

$$d_{\mathcal{R}'} \geq \sigma_{\mathcal{R}'}(\mathcal{P}). \tag{5}$$

Since $\mathcal{R}$ does not use $\mathcal{P}$, neither does $\mathcal{R}'$. This means that there is at least one arrow of $\mathcal{P}$ that is not in $\mathcal{R}'$. We know that $\mathcal{P}$ is dominant, so the shift of any such arrow is strictly larger than 0. As a result, we have $\sigma_{\mathcal{R}'}(\mathcal{P}) > 0$. Let $a = (x, y)$ be an arrow of $\mathcal{R} \setminus \mathcal{R}'$. Without loss of generality we assume $x \in T_1$ and $y \in T_2$; the symmetric case is analogous. Since $\mathcal{I}$ is an optimal $\mathcal{P}$-extension, we have $\sigma_{\mathcal{P}}(a) = 0$. In other words, we have either $\sigma(a) = 0$ or $a \in F[A_{\mathcal{P}}]$. We argue that in both cases we have $\sigma_{\mathcal{R}'}(a) < \sigma_{\mathcal{R}'}(\mathcal{P})$.

If $\sigma(a) = 0$, then we directly obtain $\sigma_{\mathcal{R}'}(a) = 0 < \sigma_{\mathcal{R}'}(\mathcal{P})$. Otherwise, if $a \in F[A_{\mathcal{P}}]$, then there exists an arrow $a' = (x', y) \in A_{\mathcal{P}}$ such that x' is a strict descendant of x. This means that the shift of a' is strictly greater than the shift of a. We consider two cases. First, if $a' \in A_{\mathcal{R}'}$ then a is in the fan of $A_{\mathcal{R}'}$ and we obtain $\sigma_{\mathcal{R}'}(a) = 0 < \sigma_{\mathcal{R}'}(\mathcal{P})$. Otherwise, if $a' \notin A_{\mathcal{R}'}$, then $\sigma_{\mathcal{R}'}(a') = \sigma(a')$. Hence, we obtain $\sigma_{\mathcal{R}'}(\mathcal{P}) \geq \sigma(a') > \sigma(a) \geq \sigma_{\mathcal{R}'}(a)$. In all cases, we obtain $\sigma_{R'}(a) < \sigma_{\mathcal{R}'}(\mathcal{P})$. Combining with Equation (5), we get $\sigma_{\mathcal{R}'}(a) < d_{\mathcal{R}'}$. By Corollary 23, it then follows that $d_{\mathcal{R}} \geq d_{\mathcal{R}'}$. It remains to show that $\sigma_{\mathcal{R}'}(\mathcal{P}) \geq \sigma_{\mathcal{R}}(\mathcal{I})$. Since any arrow of $\mathcal{R} \setminus \mathcal{R}'$ is not an arrow of $\mathcal{P}$, we have $\sigma_{\mathcal{R}'}(\mathcal{P}) = \sigma_{\mathcal{R}}(\mathcal{P})$. Consider an arrow $a = (x, y)$ of $\mathcal{I}$. Again, without loss of generality assume $x \in T_1$ and $y \in T_2$; the symmetric case is analogous. The $\mathcal{P}$-shift of any arrow a is 0, so either $\sigma(a) = 0$, or $a \in F[A_{\mathcal{P}}]$. In the first case, we get $\sigma_{\mathcal{R}}(a) = 0 < \sigma_{\mathcal{R}}(\mathcal{P})$. In the second case, there must be an arrow $a' = (x', y)$ of $\mathcal{P}$ with $x' \preceq x$. It follows that the shift of a' is at least as great as the shift of a, and thus $\sigma_{\mathcal{R}}(a) \leq \sigma_{\mathcal{R}}(a') \leq \sigma_{\mathcal{R}}(\mathcal{P})$. Since this holds for all arrows of $\mathcal{I}$, we get $\sigma_{\mathcal{R}}(I) \leq \sigma_{\mathcal{R}}(\mathcal{P}) \leq \sigma_{\mathcal{R}'}(\mathcal{P})$. This concludes the proof. □

4.3 Computing a Locally Correct Interleaving

We sketch how the proof of Theorem 19 can be translated into an incremental algorithm for computing a locally correct interleaving. To compute the residual interleaving distance, we modify the dynamic program by Touli and Wang [13] to also incorporate constraints induced by partial interleavings. We use the output to extract a finite set of critical pairs, that corresponds to an augmentation. We then iteratively reduce the set of critical pairs by repeatedly calling the modified dynamic program until we obtain a (finite) minimal augmentation. This minimal augmentation, in turn, reduces the residual interleaving distance; we iterate until it has dropped to zero. To obtain a complete locally correct interleaving, we select suitable entries from the table filled by the modified dynamic program.

5 Conclusion

We introduced locally correct interleavings between merge trees, which are optimal interleavings that are "tight". For this, we introduced the residual interleaving distance: a generalized version of the interleaving distance that can accommodate constraints on the interleavings. We presented a constructive proof that there always exists a locally correct interleaving. We plan to further investigate the complexity of computing a locally correct interleaving. In particular, we will study the effect of partial interleavings on the running time of the modified dynamic program. Furthermore, we will explore the effects of replacing the dynamic program with the approximation algorithm [1] or one of the variants or heuristics. Moreover, it would be interesting to study other (local) criteria of interleavings. Since locally correct interleavings are not necessarily unique, we could minimize their total shift, for a suitable definition of total. Furthermore, we could also consider a lexicographic optimization, inspired by lexicographic Fréchet matchings [11]. Lastly, we want to explore in which way locally correct interleavings can be translated to meaningful matchings between terrains.

References

[1] P.K. Agarwal, K. Fox, A. Nath, A. Sidiropoulos, and Y. Wang. Computing the Gromov-Hausdorff distance for metric trees. ACM Transactions on Algorithms, 14(2):1–20, 2018.

[2] T. Beurskens, T. Ophelders, B. Speckmann, and K. Verbeek. Relating interleaving and Fréchet distances via ordered merge trees. In Proc. ACM-SIAM Symposium on Discrete Algorithms (SODA25), pages 5027–5050. Society for Industrial and Applied Mathematics, 2025.

[3] K. Buchin, M. Buchin, W. Meulemans, and B. Speckmann. Locally correct Fréchet-matchings. Computational Geometry, 76(1):1–18, 2019.

[4] J. Curry, H. Hang, W. Mio, T. Needham, and O. B. Okutan. Decorated merge trees for persistent topology. Journal of

Applied and Computational Topology, 6(3):371–428, February 2022.

[5] E. Gasparovic, E. Munch, S. Oudot, K. Turner, B. Wang, and Y. Wang. Intrinsic interleaving distance for merge trees. La Matematica, 4(1):40–65, 2025.

[6] E. Gasparovich, E. Munch, S. Oudot, K. Turner, B. Wang, and Y. Wang. Intrinsic interleaving distance for merge trees.

[7] D. Morozov, K. Beketayev, and G. Weber. Interleaving distance between merge trees. Manuscript (accessed on 06-03-2025), 2013.

[8] E. Munch and A. Stefanou. The ℓ^∞-Cophenetic metric for phylogenetic trees as an interleaving distance. In Research in Data Science, volume 17 of Association for Women in Mathematics Series, pages 109–127. Springer International Publishing, 2019.

[9] M. Pegoraro. A graph-matching formulation of the interleaving distance between merge trees. AIMS Mathematics, 10(6):13025–13081, 2025.

[10] M. Pont, J. Vidal, J. Delon, and J. Tierny. Wasserstein distances, geodesics and barycenters of merge trees. IEEE Transactions on Visualization and Computer Graphics, 28(1):291–301, January 2022.

[11] G. Rote. Lexicographic Fréchet matchings. In Proc. 30th European Workshop on Computational Geometry (EuroCG'14), 2014.

[12] R. Sridharamurthy, T. B. Masood, A. Kamakshidasan, and V. Natarajan. Edit distance between merge trees. IEEE Transactions on Visualization and Computer Graphics, 26(3):1518–1531, 2020.

[13] E.F. Touli and Y. Wang. FPT-algorithms for computing the Gromov-Hausdorff and interleaving distances between trees. Journal of Computational Geometry, 13:89–124, 2022.

[14] L. Yan, T. B. Masood, F. Rasheed, I. Hotz, and B. Wang. Geometry aware merge tree comparisons for time-varying data with interleaving distances. IEEE Transactions on Visualization and Computer Graphics, 29(8):3489–3506, 2022.

[15] L. Yan, T. B. Masood, R. Sridharamurthy, F. Rasheed, V. Natarajan, I. Hotz, and B. Wang. Scalar field comparison with topological descriptors: Properties and applications for scientific visualization. Computer Graphics Forum, 40(3):599–633, 2021.

[16] L. Yan, Y. Wang, E. Munch, E. Gasparovich, and B. Wang. A structural average of labeled merge trees for uncertainty visualisation. IEEE Transactions on Visualization and Computer Graphics, 26(1):832–842, 2020.